

Data science lifecycle & Exploratory data analysis using visualization

ds4owd - data science for openwashdata

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ETH Zurich

Sep 18, 2025

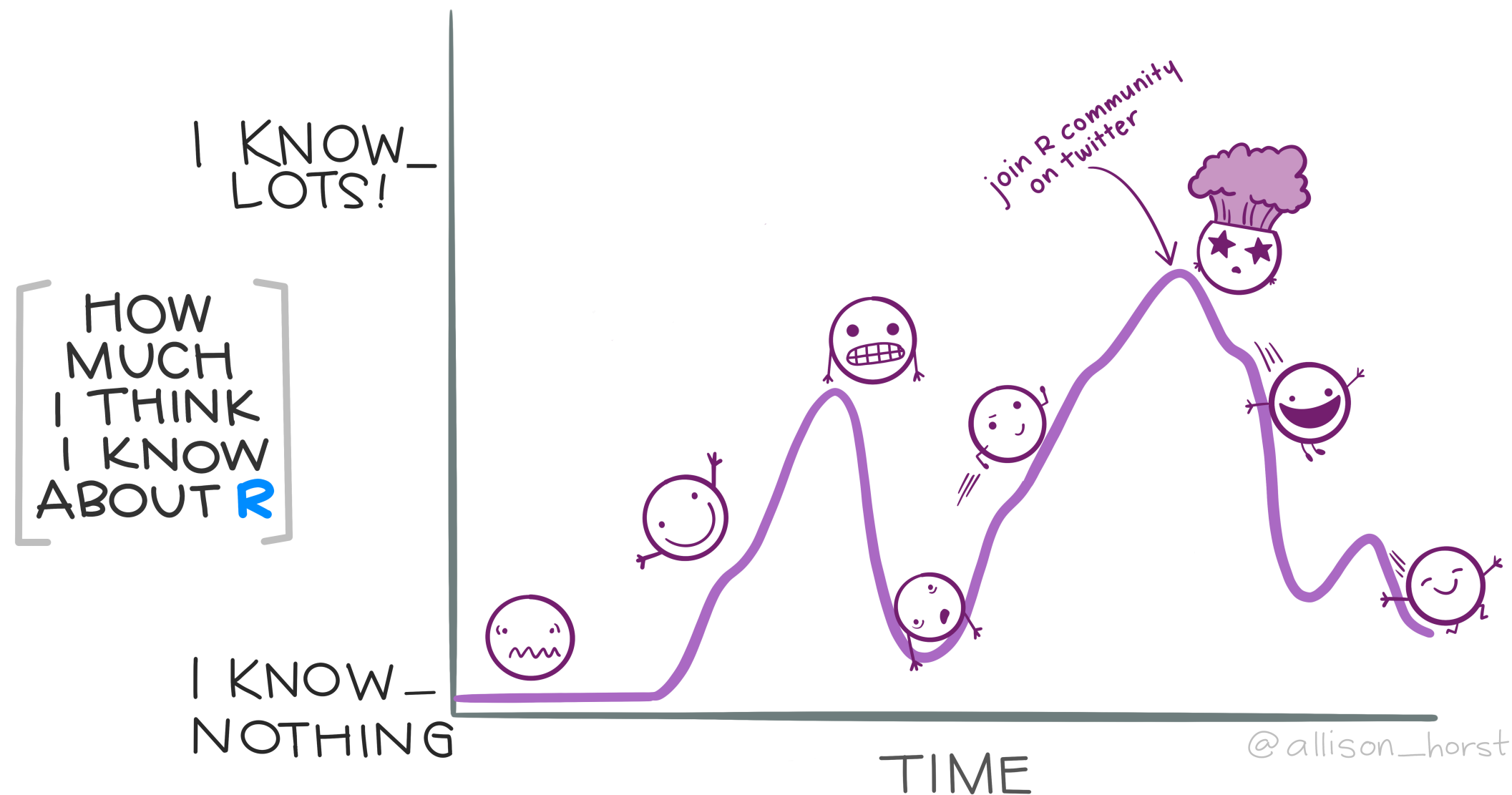


Q: How do I successfully complete the course?

You successfully complete the course and you will receive a certificate of completion if you:

1. Complete the quiz of each week by its due date
- 2- Hand in a complete capstone project report that uses a dataset of your choice by 11 December 2025 (instructions will follow)

While homework assignments are not required for completion, we highly recommend working on them and submitting it for



Solving coding problems online

Tips for search engines (e.g. duckduckgo.com)

- Use actionable verbs that describe what you want to do
- Be specific
- Add R to the search query
- Add the name of the R package name to the search query
- Scroll through the top 5 results (don't just pick the first)

Example: “How to remove a legend from a plot in R ggplot2”

Stack Overflow

What is it?

- The biggest support network for (coding) problems
- Can be intimidating at first
- Up-vote system

Workflow

- First, briefly read the question that was posted
- Then, read the answer marked as “correct”
- Then, read one or two more answers with high votes

Tips for AI tools (e.g. duck.ai)

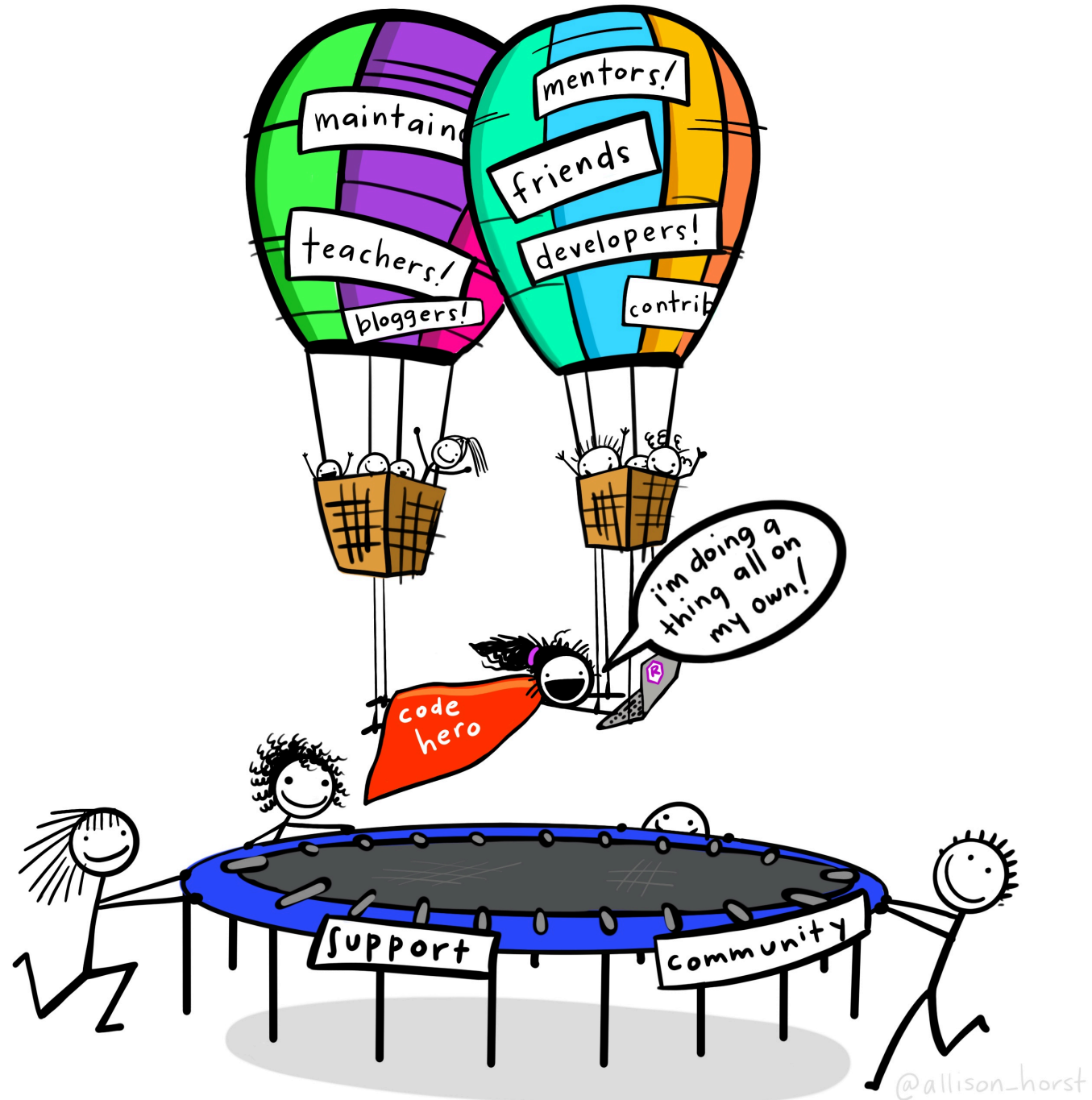
- Use actionable verbs that describe what you want to do
- Be specific
- Add R to the search query
- Add the name of the R package name to the search query

Example: “How to remove a legend from a plot in R ggplot2”

Other sources for help

- Posit Community Forum:
<https://community.rstudio.com/>
- Documentation websites:
<https://ggplot2.tidyverse.org/>
- Mastodon / Bluesky tag:
[#rstats](#)





Exercises & Assignments

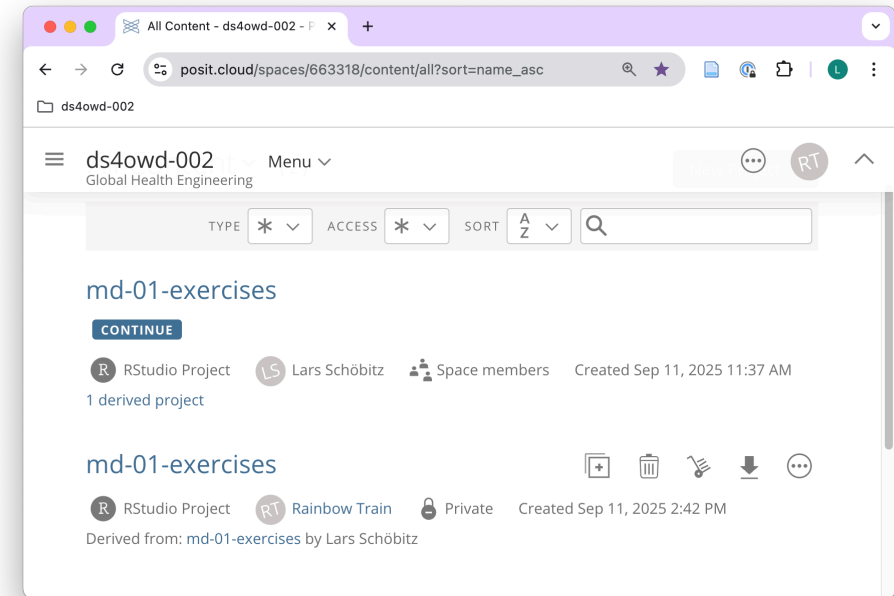
Note

For every week, there will be two “projects”:

- `md-0X-exercises` (this is for lectures and we only work with it during lectures)
- `md-0X-assignments-USERNAME` (this is your homework and every module will have your username at the end of it. You will clone this to Posit Cloud to work with it)

Lecture Exercises

- Click **Start** the first time
- Click **Continue** the next time
- It will look like two projects, but it is just one
- It doesn't matter on which of the two you click



Homework Assignments

on GitHub Organisation

Assignment 1 of Module 2 is a Bookmark Folder assignment!

<https://github.com/ds4owd-002>

ds4owd-001

github.com/ds4owd-001?q=rainbow-train&type=all&language=&sort=

website Public

Forked from [cven5873-ss23/website](#)

Website for data science for openwashdata course 001

JavaScript

Top languages

HTML JavaScript

Repositories

rainbow-train

Type Language Sort New

1 result for all repositories matching rainbow-train sorted by last updated

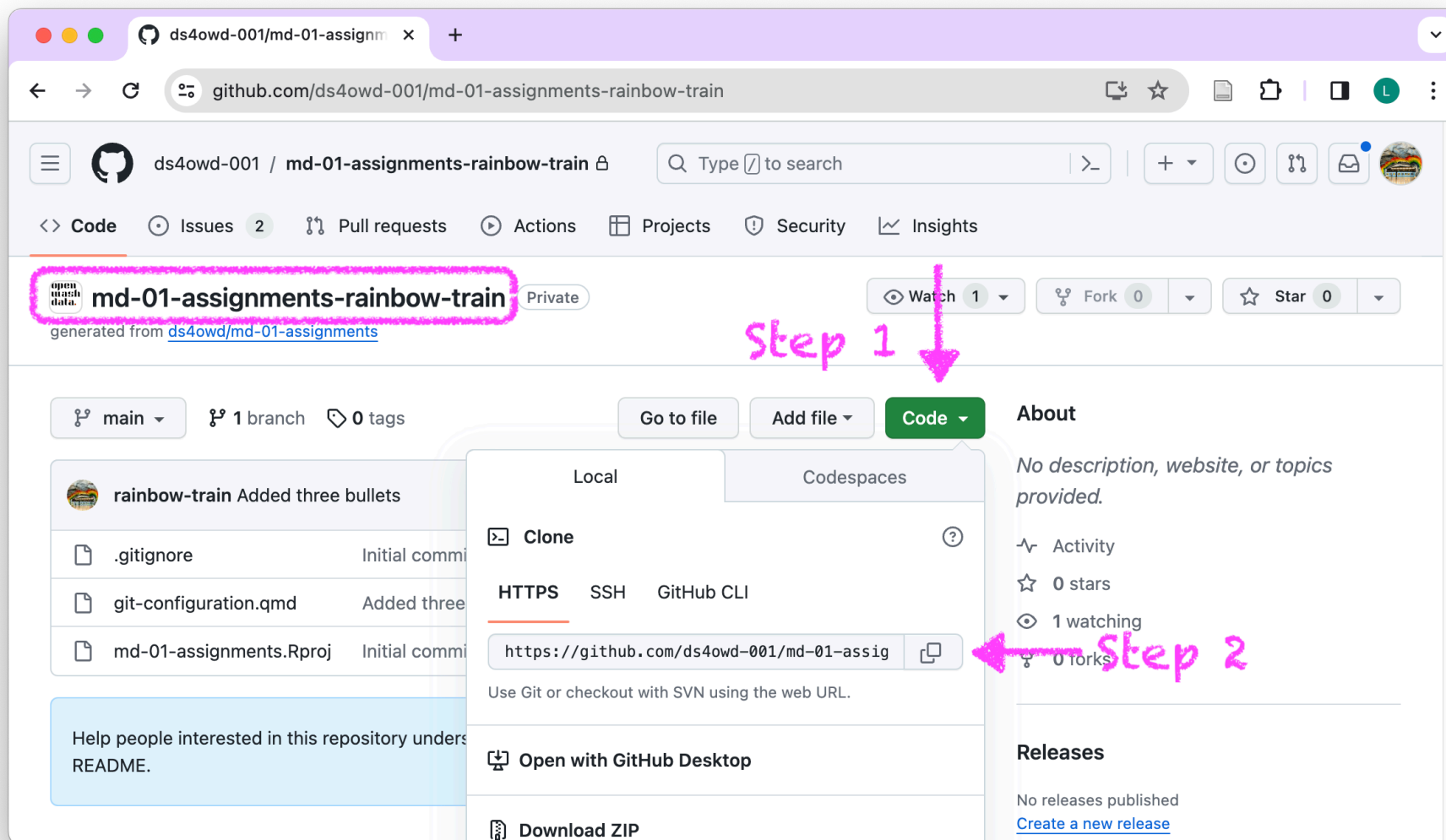
Clear filter

md-01-assignments-rainbow-train Private

0 0 2 0 Updated 4 days ago

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on your repository



on Posit Cloud

posit.cloud/spaces/663318/

Posit Cloud

posit.cloud/spaces/426916/content/all?sort=name_asc

ds4owd-001
Global Health Engineering

All Content (3)

TYPE * ACCESS * SORT

md-01-assignments-rainbow-train

RStudio Project Rainbow Train Private Created Nov 2, 2023 2:5

md-01-exercises

RStudio Project Lars Schöbitz Space members Created Oct 31, 2023 11:18 AM 1 derived project

md-01-exercises

RStudio Project Rainbow Train Private Created Oct 31, 2023 2:33 PM

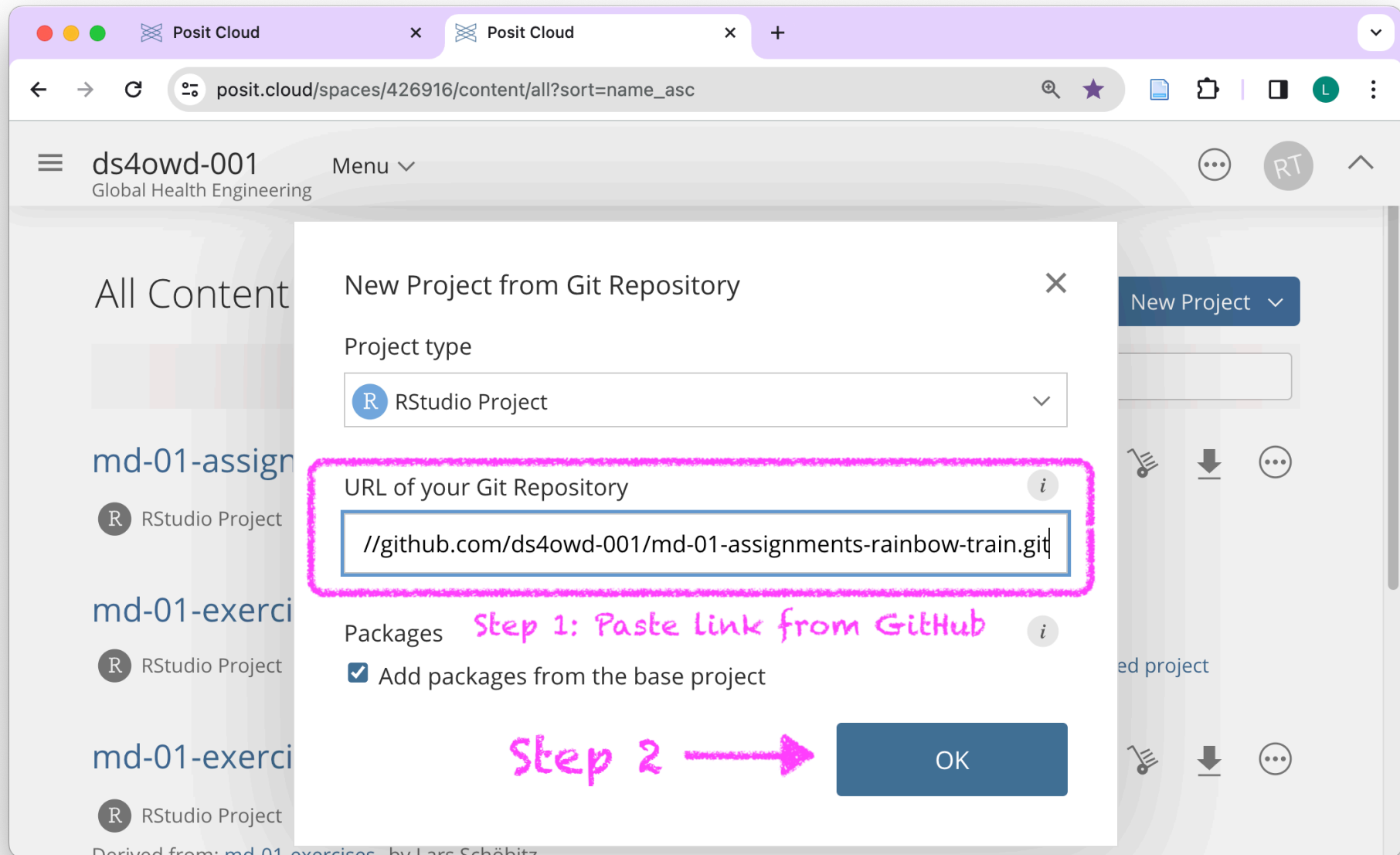
Derived from: md-01-exercises by Lars Schöbitz

Step 1 → New Project

- New Project from Template
- New RStudio Project
- New Jupyter Project
- New Project from Git Repository

Step 2 →

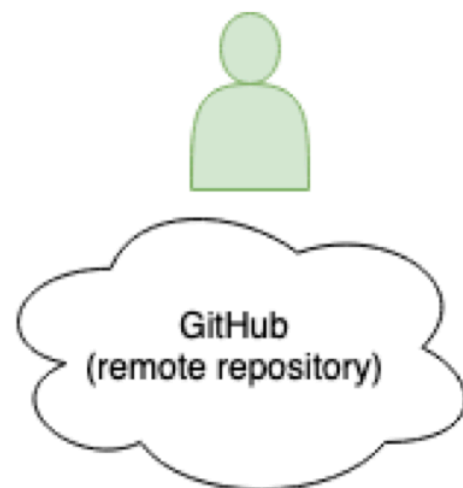
on Posit Cloud

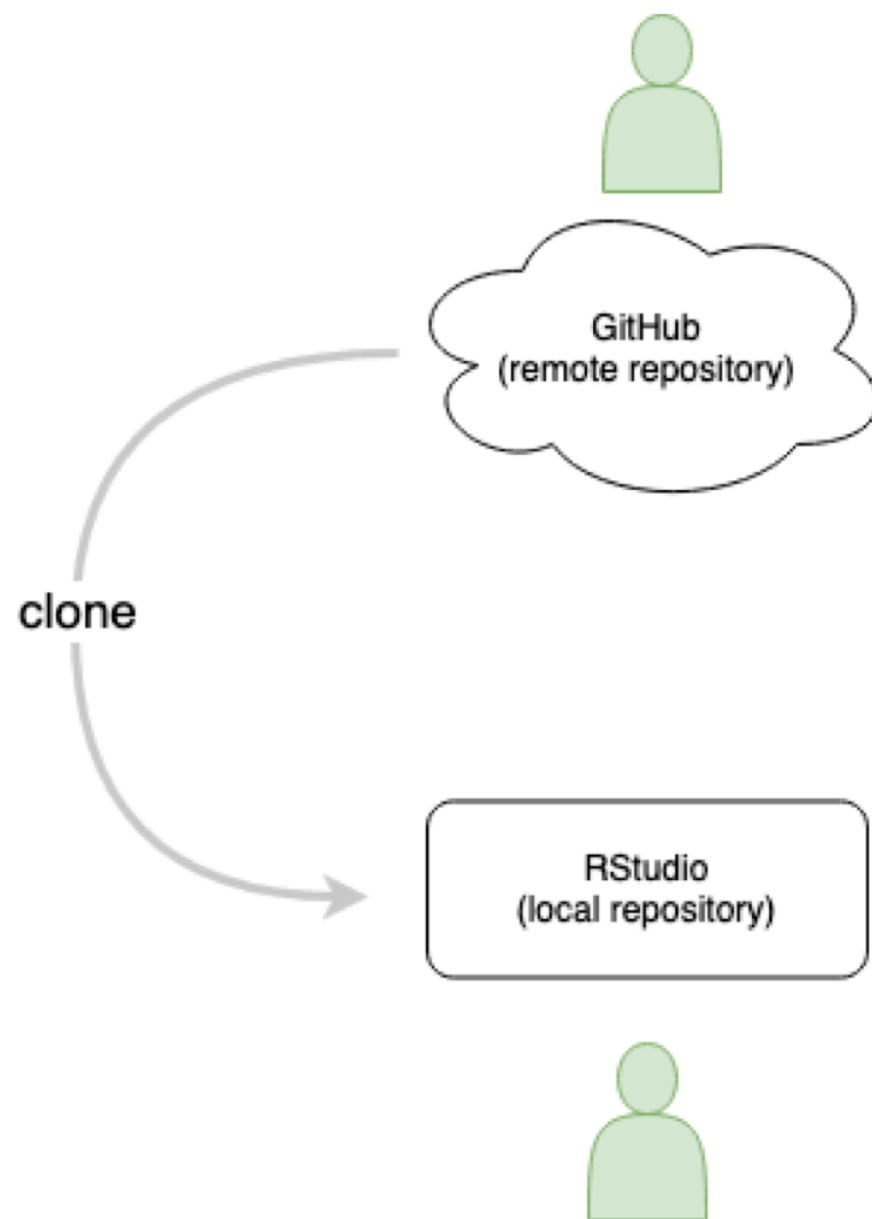


Module 2 - Assignment 1: Bookmarks

This week, you will have an assignment for creating bookmarks in your browser. It is very useful and will support you through the course and beyond.

Version Control - Terminology







Create README.md

75aa637

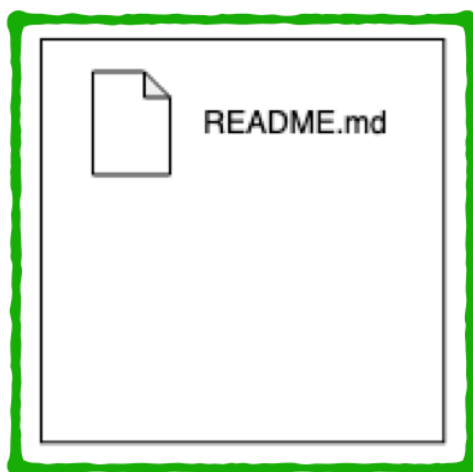


README.md

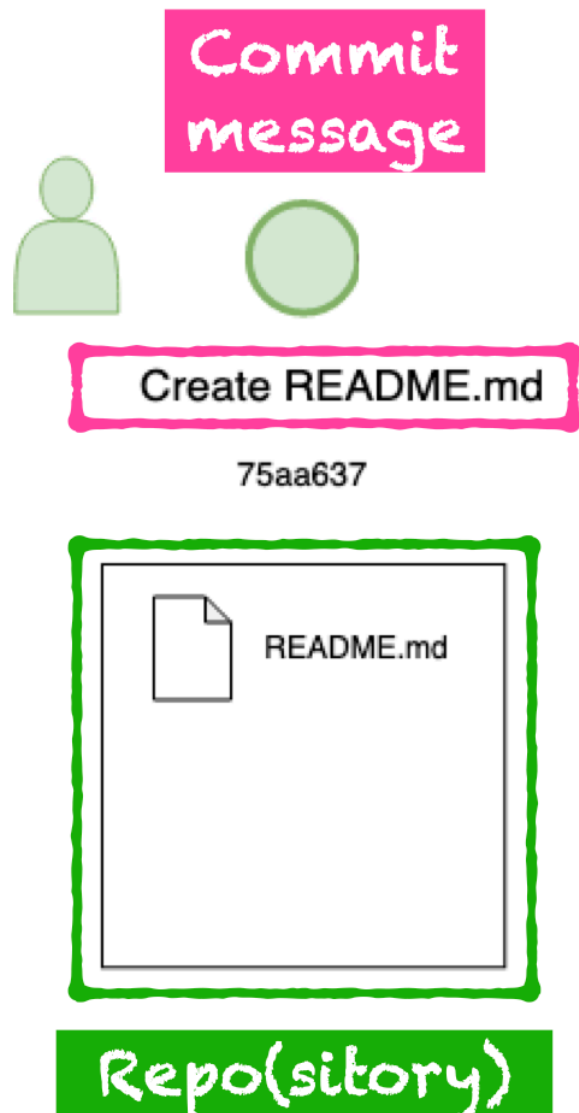


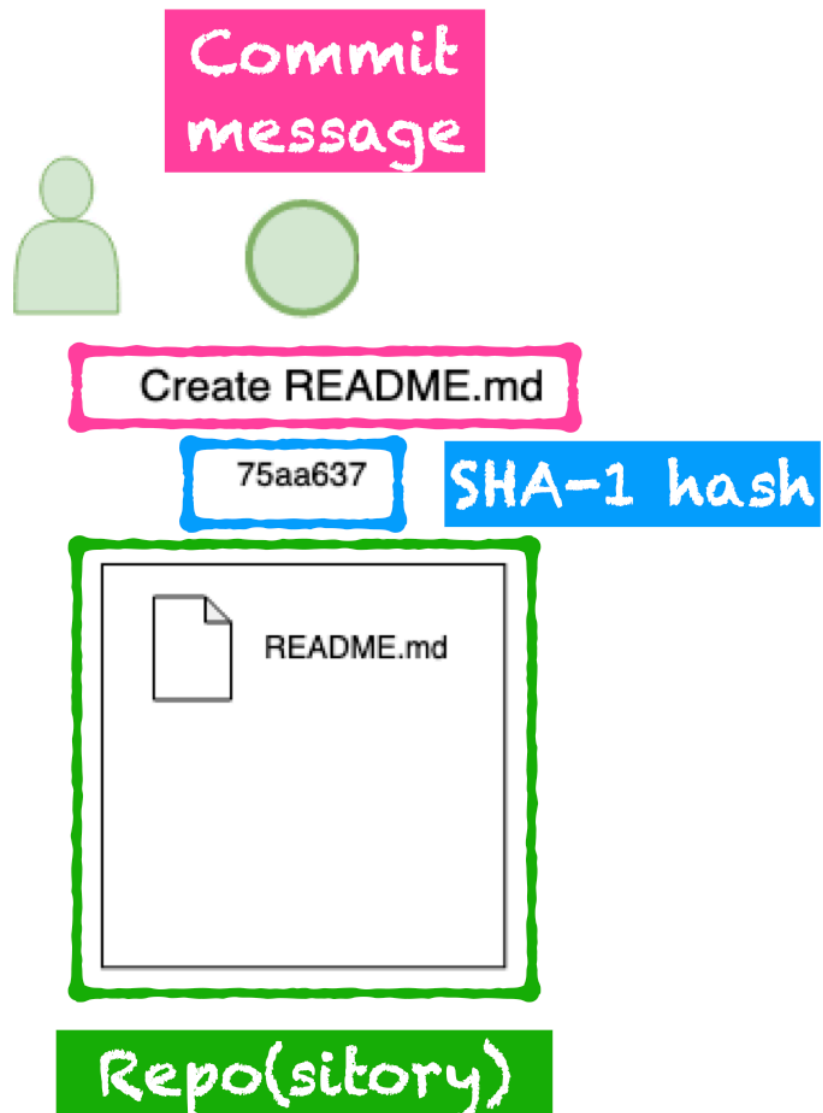
Create README.md

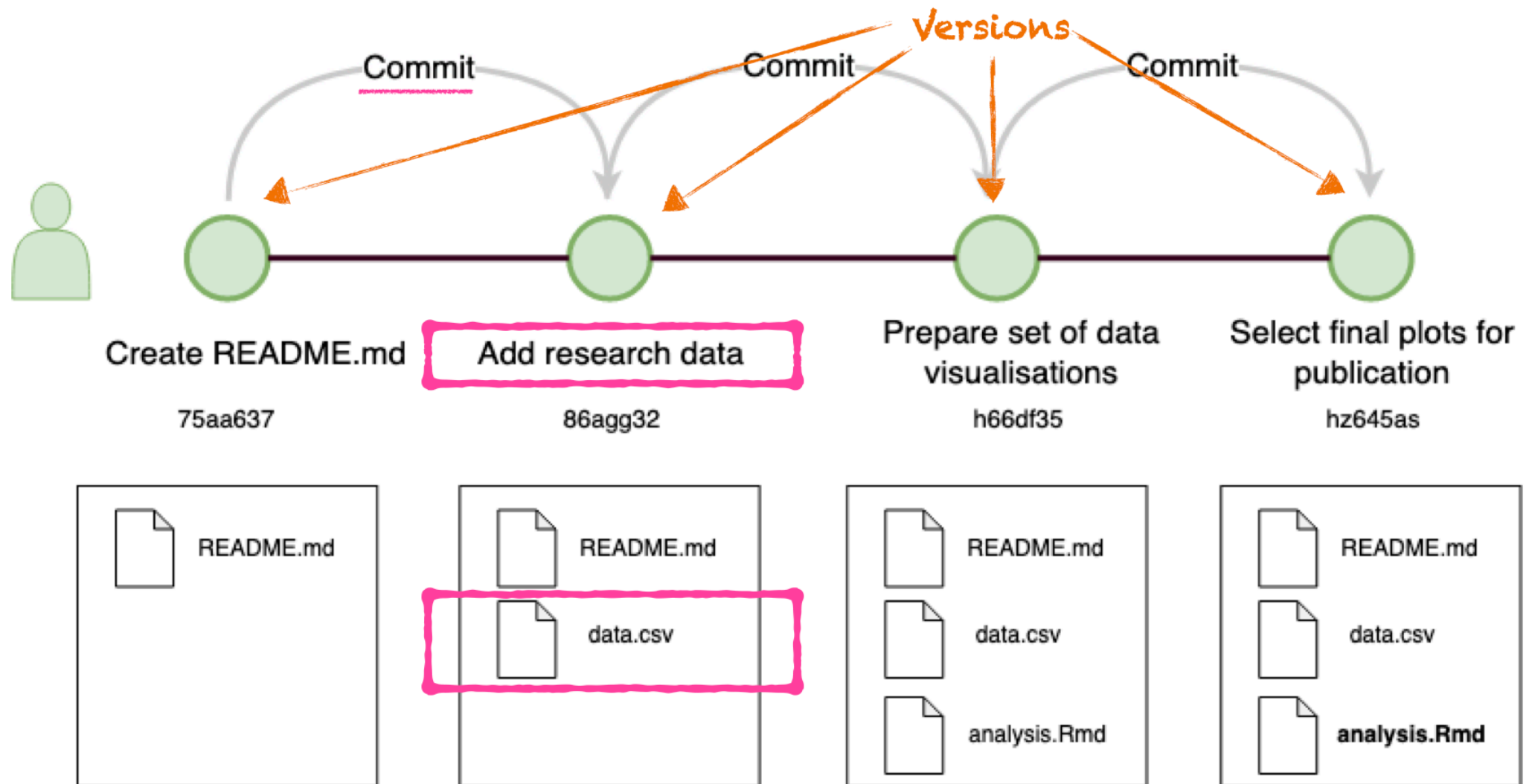
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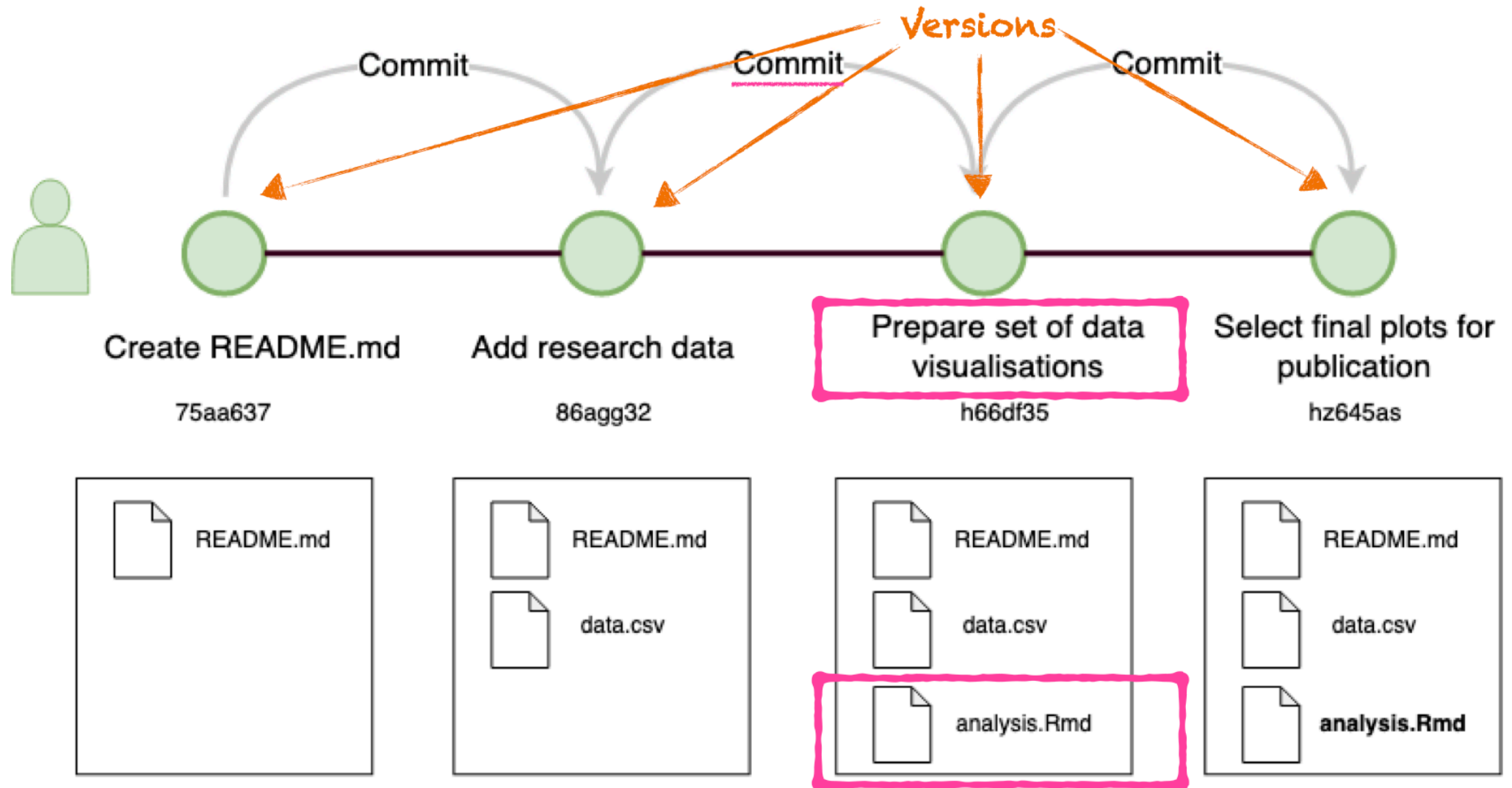


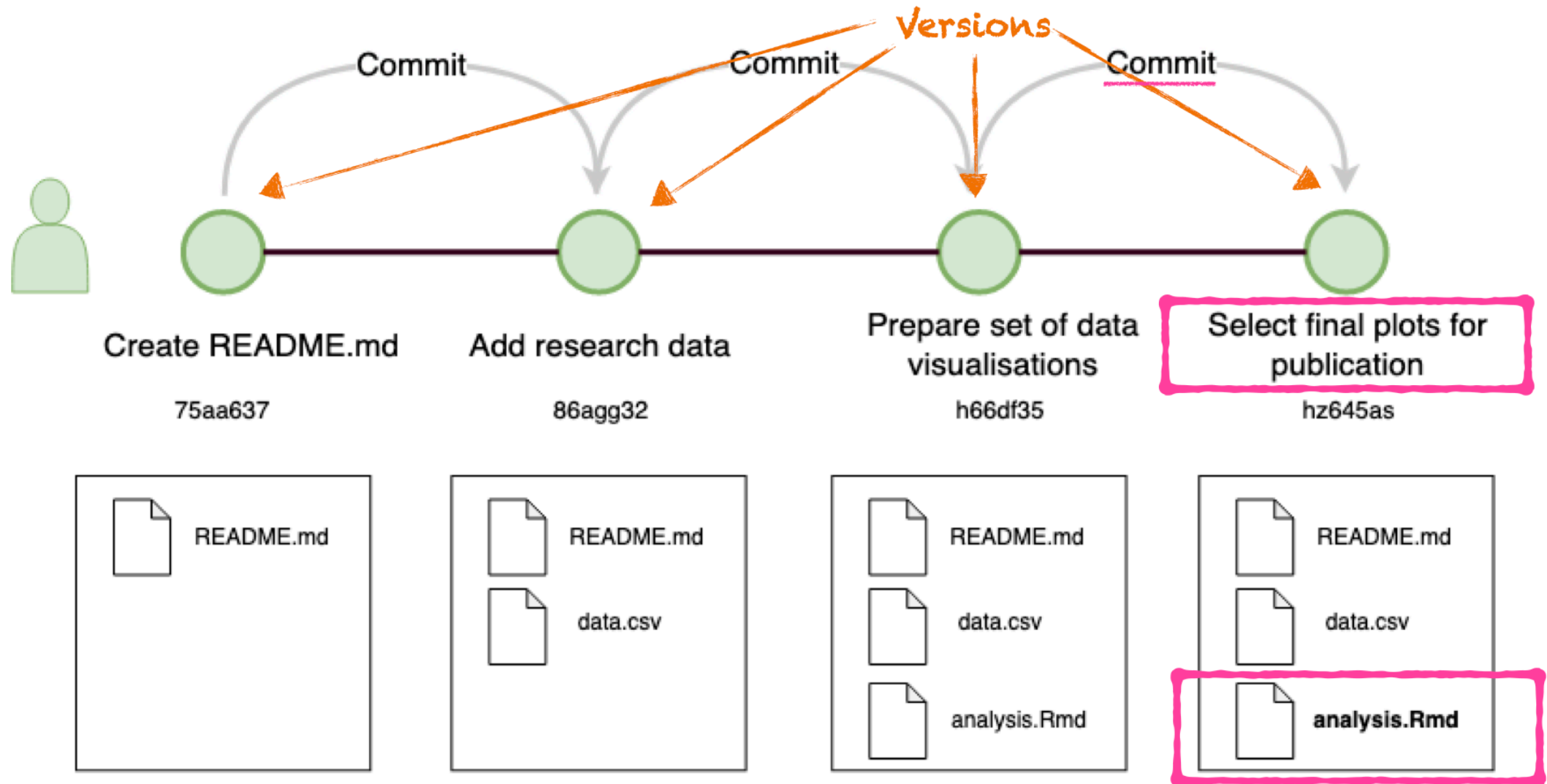
Repo(sitory)

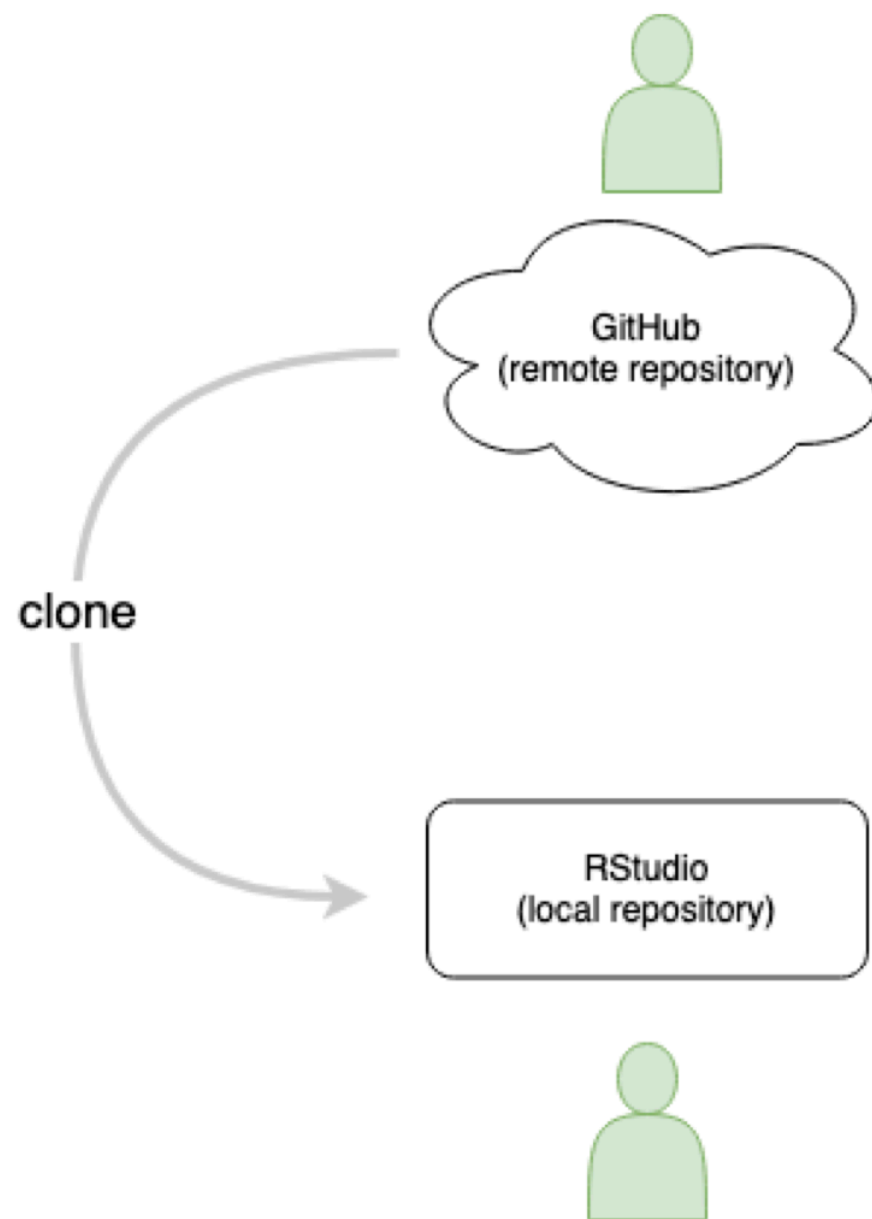


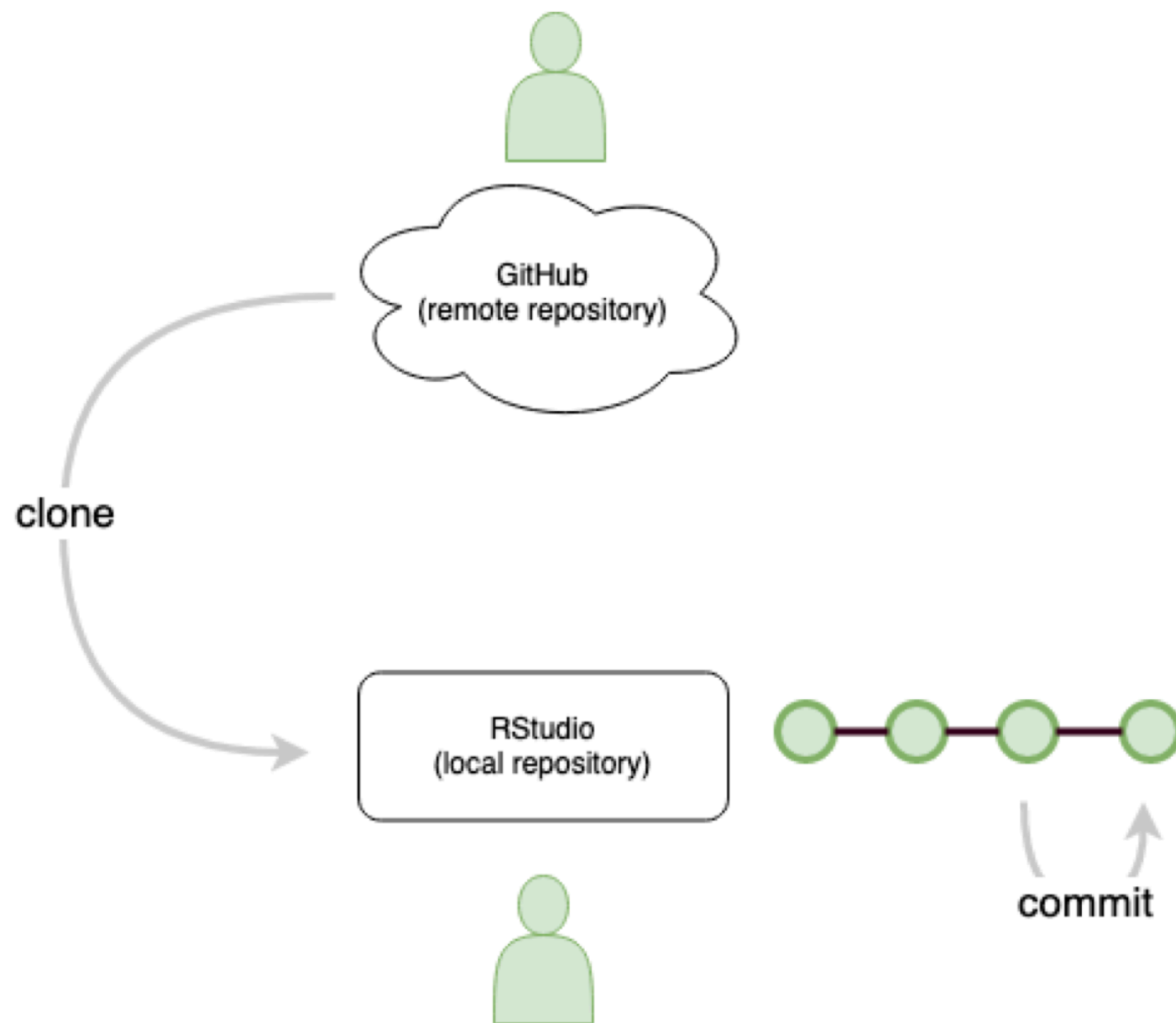


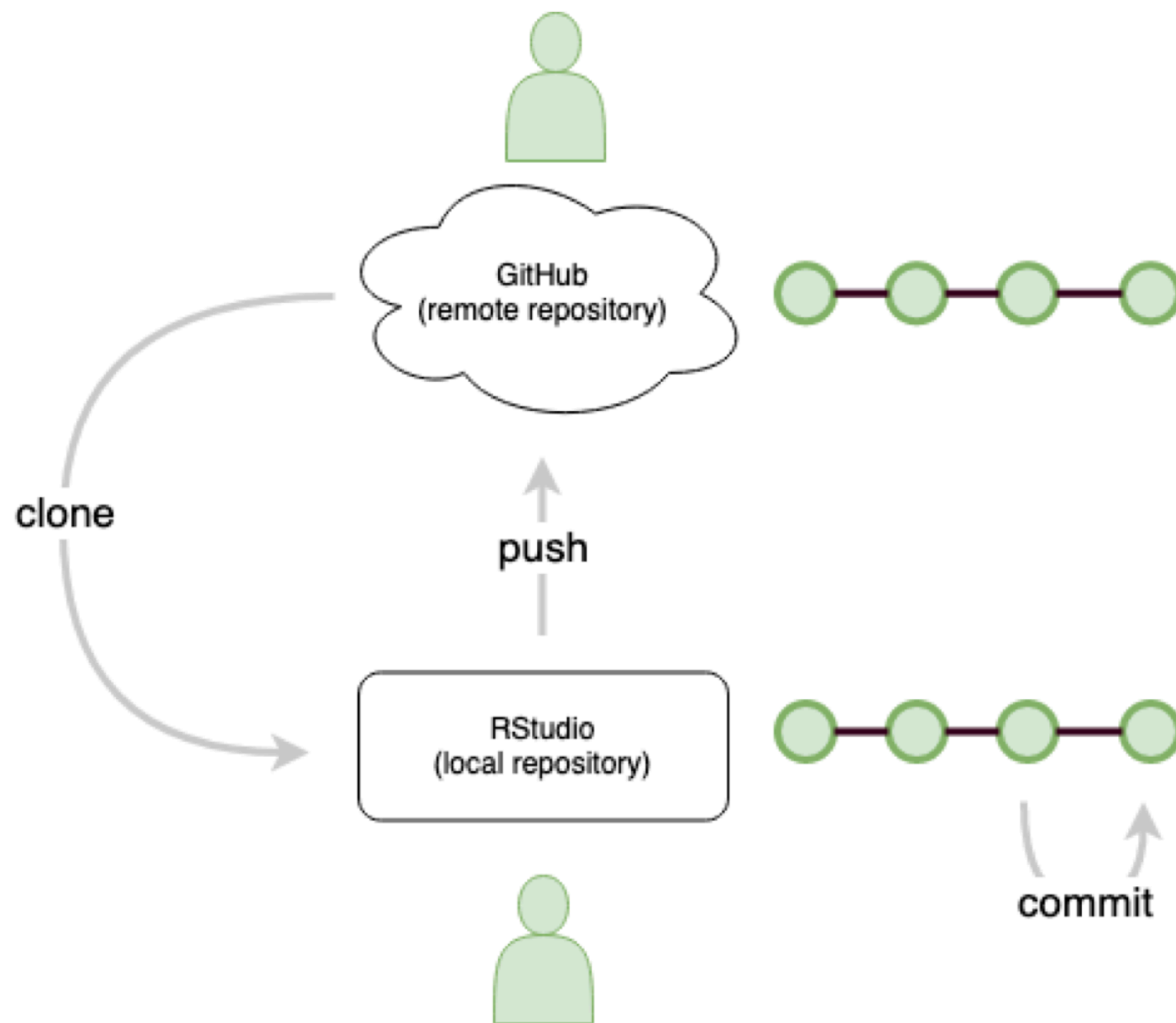




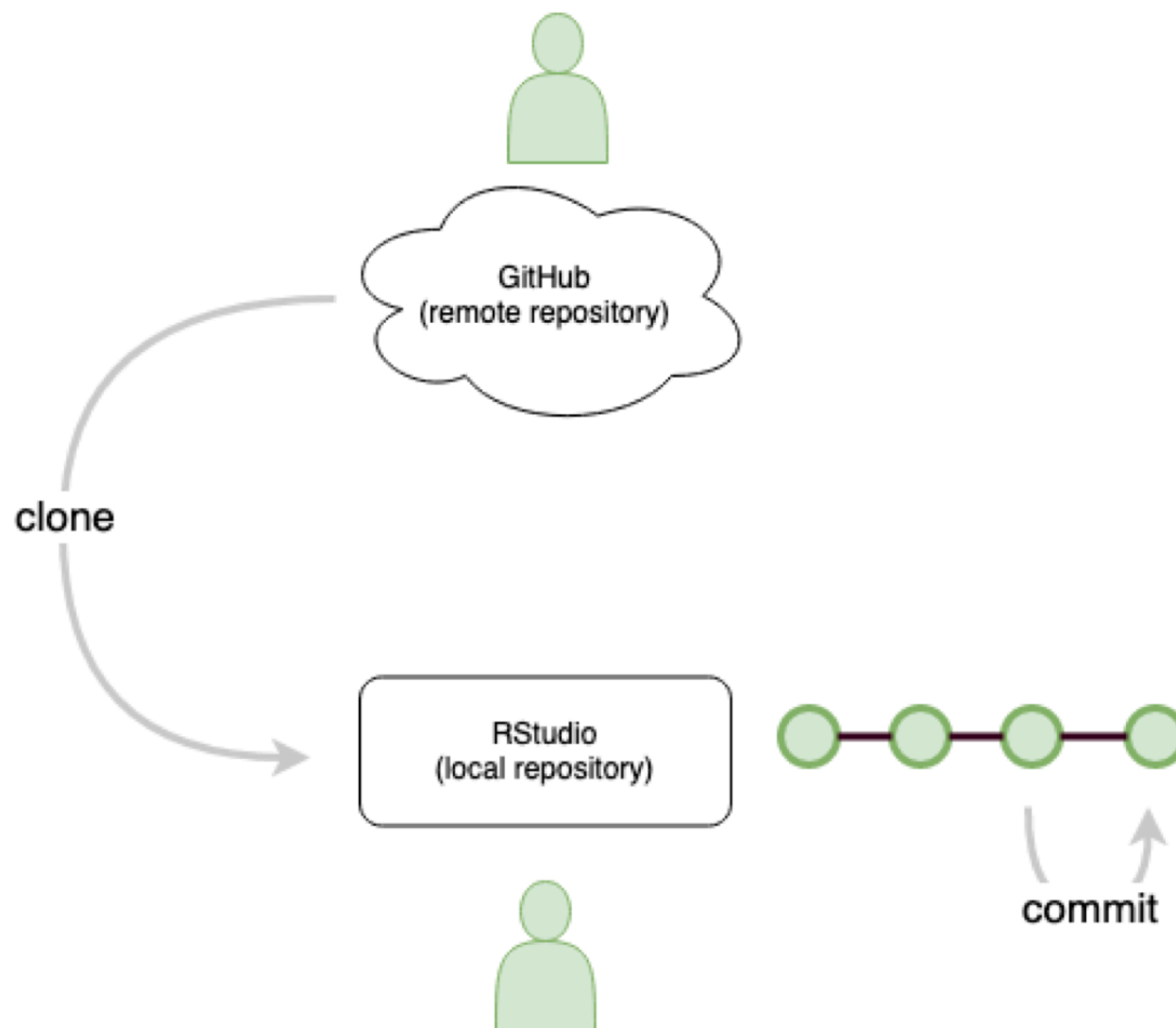




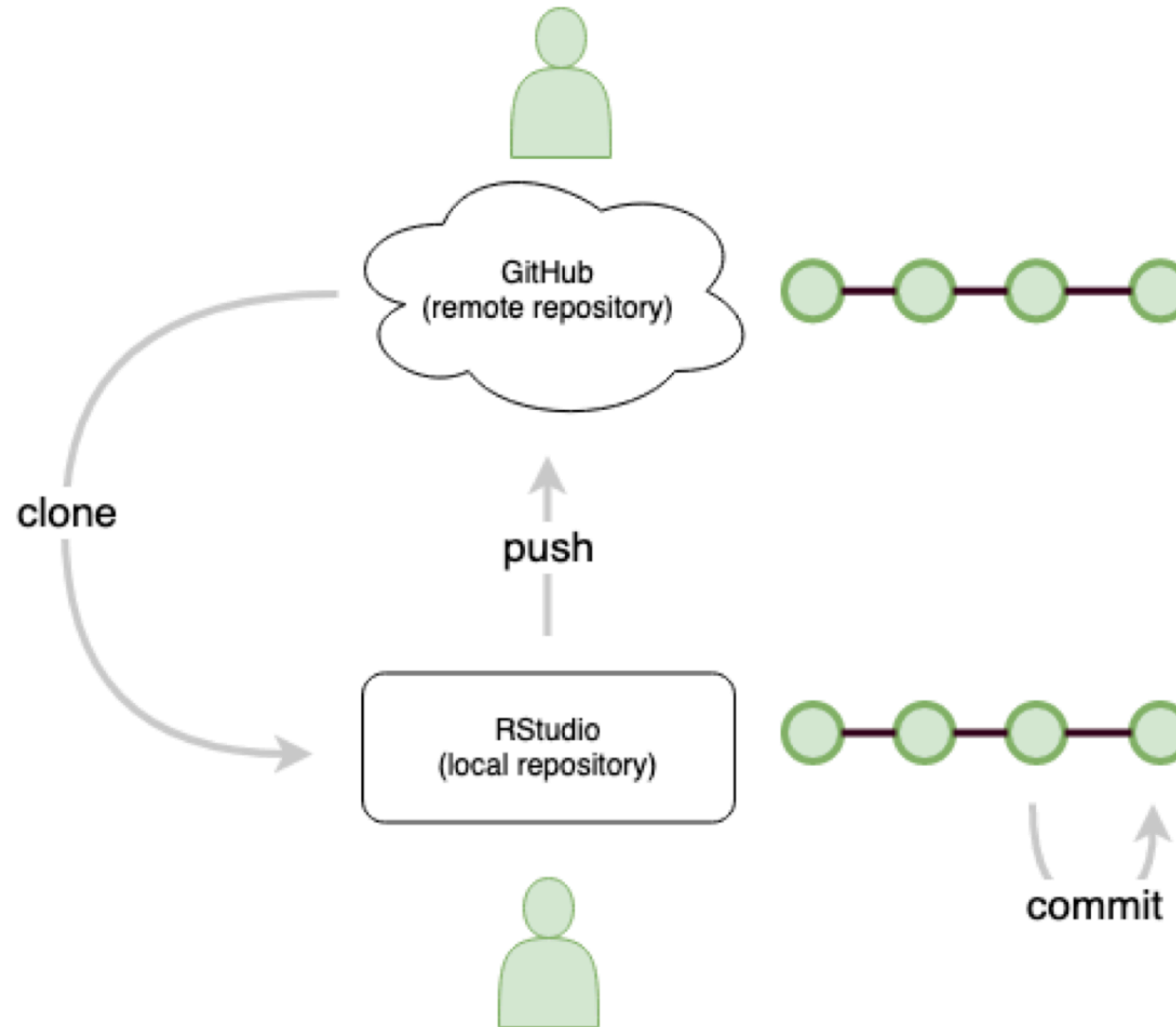




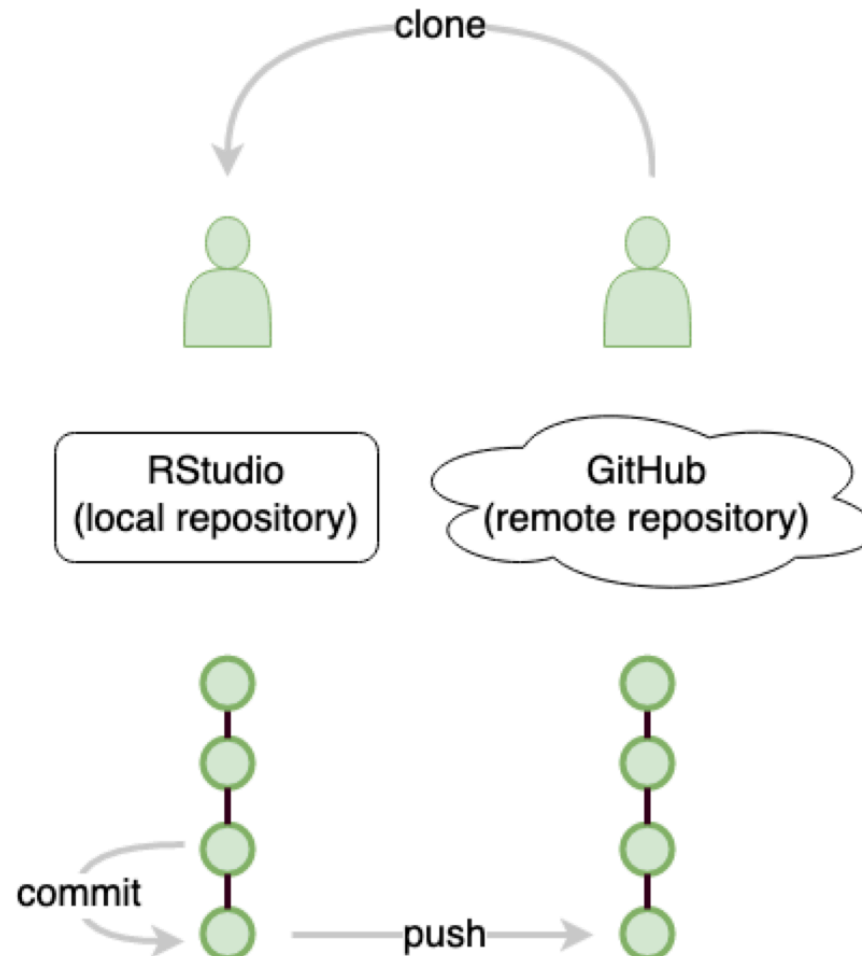
remember: git commit



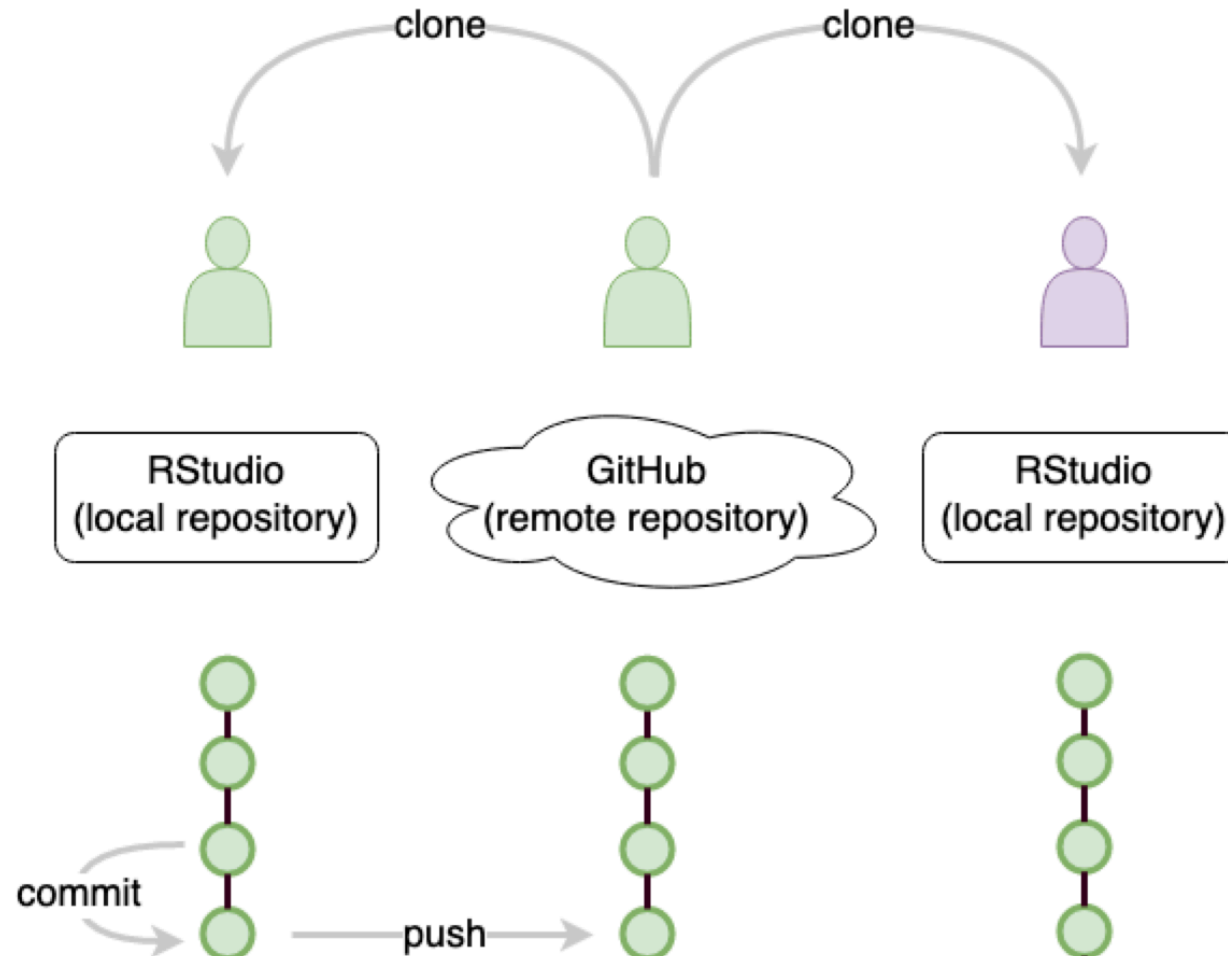
remember: git push



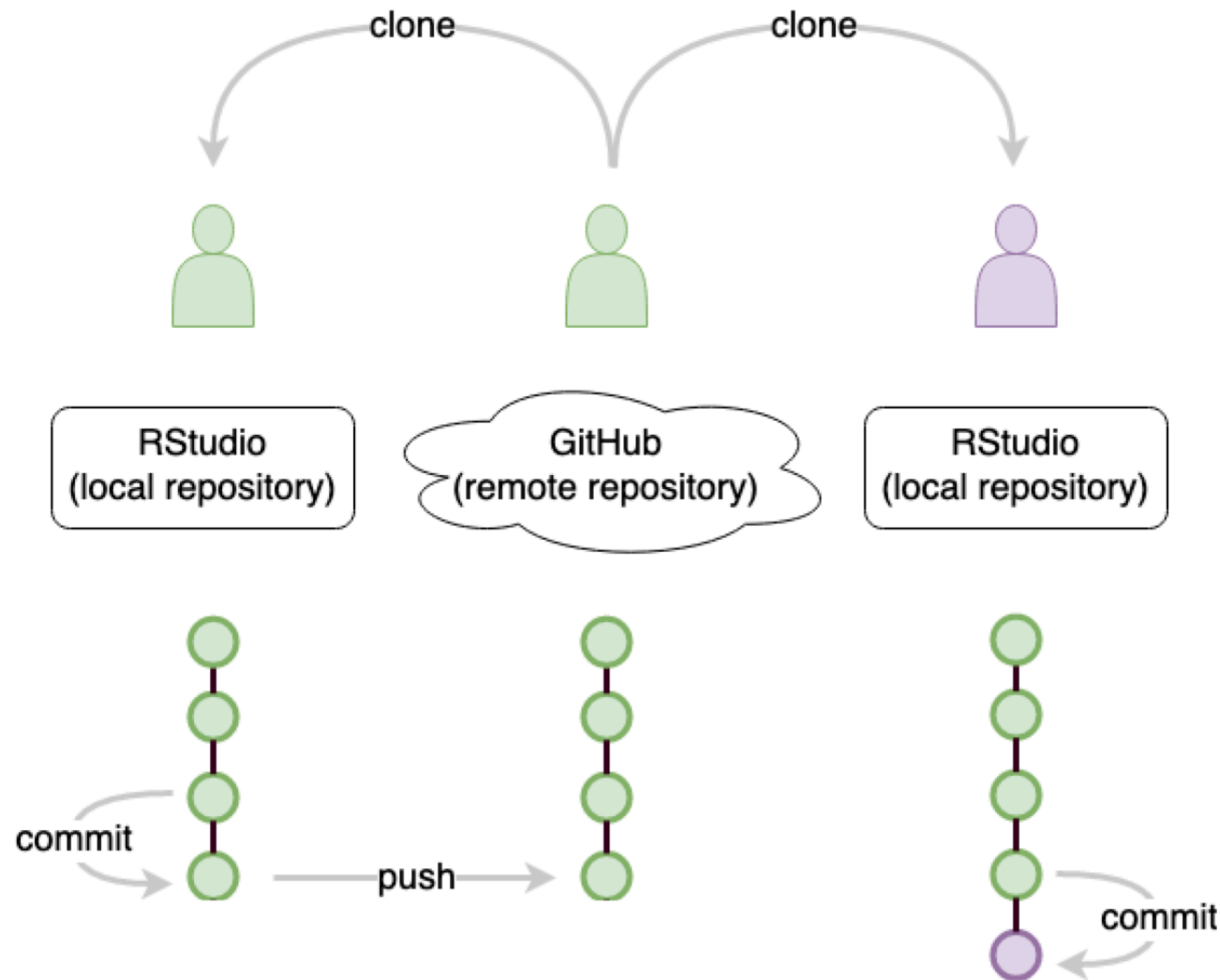
remember: `git push`



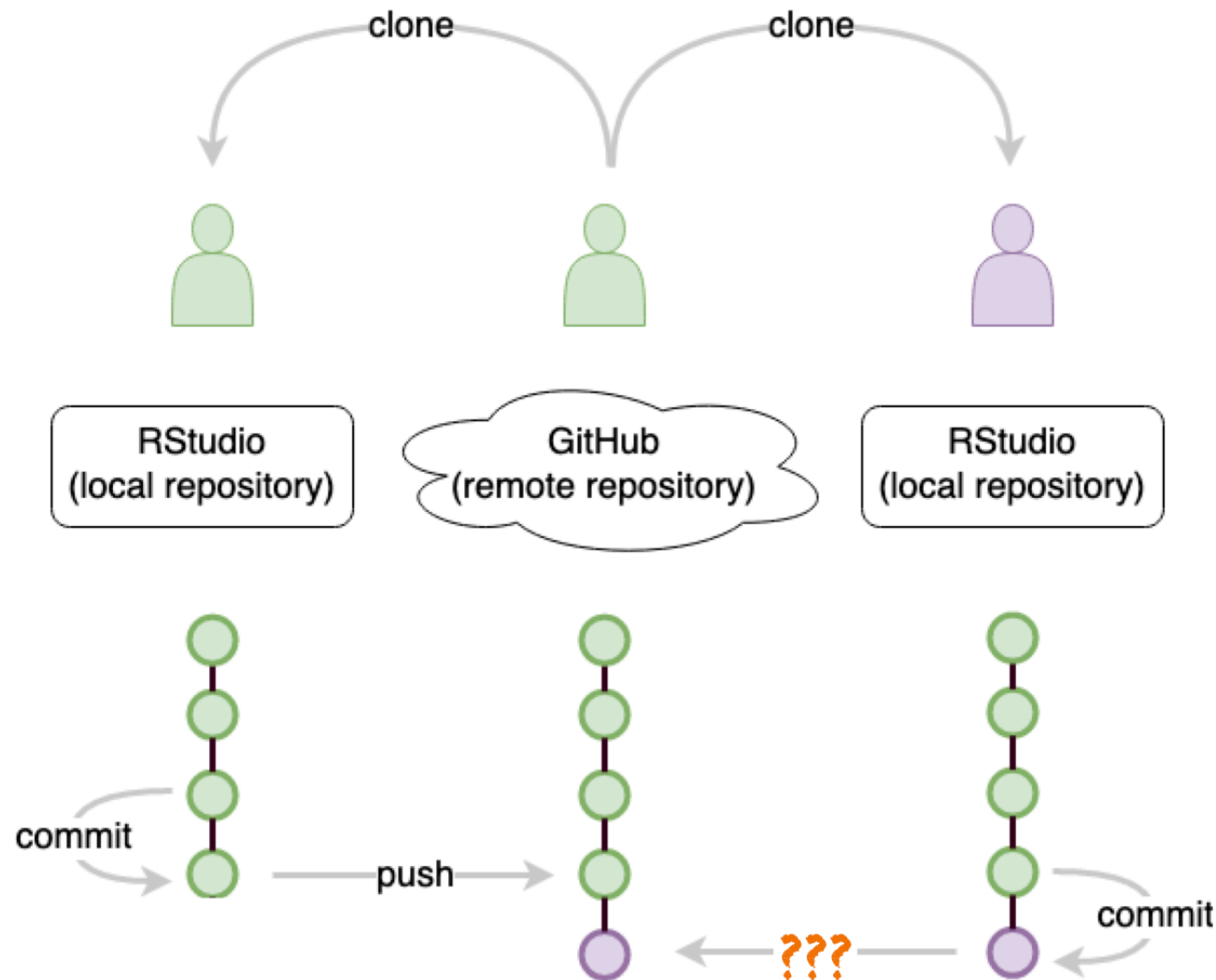
collaborate: `git clone`



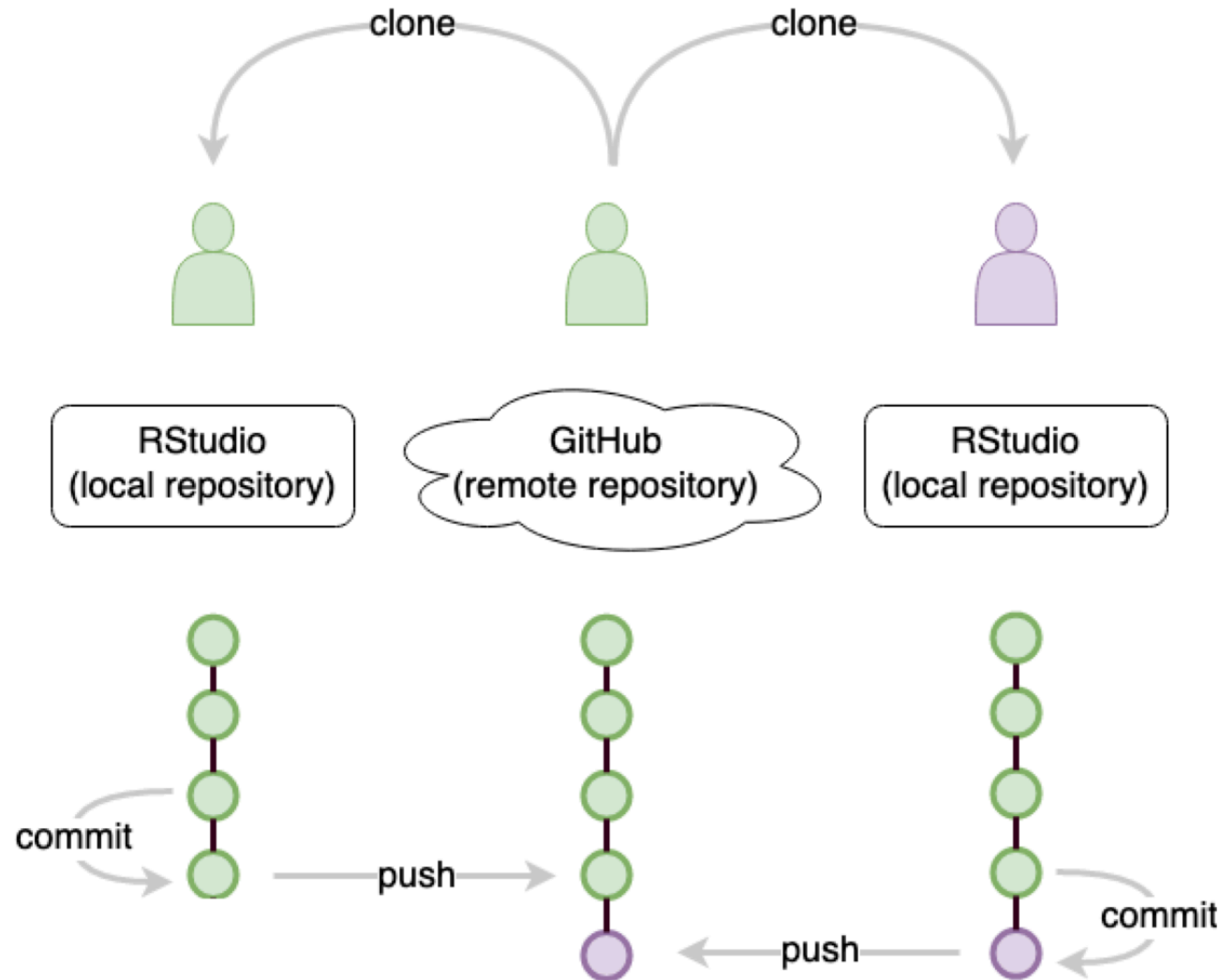
track work: git commit



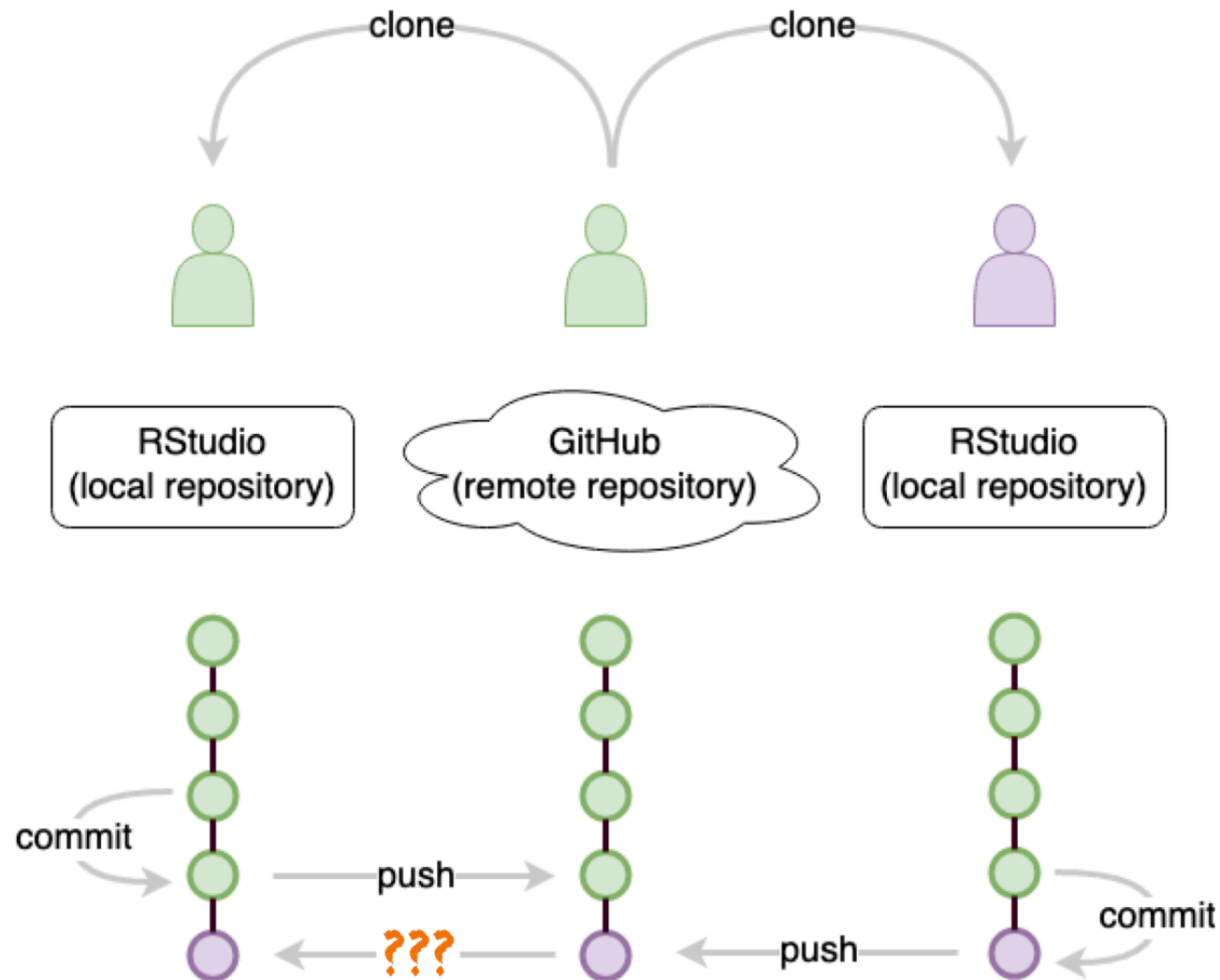
update: git ???



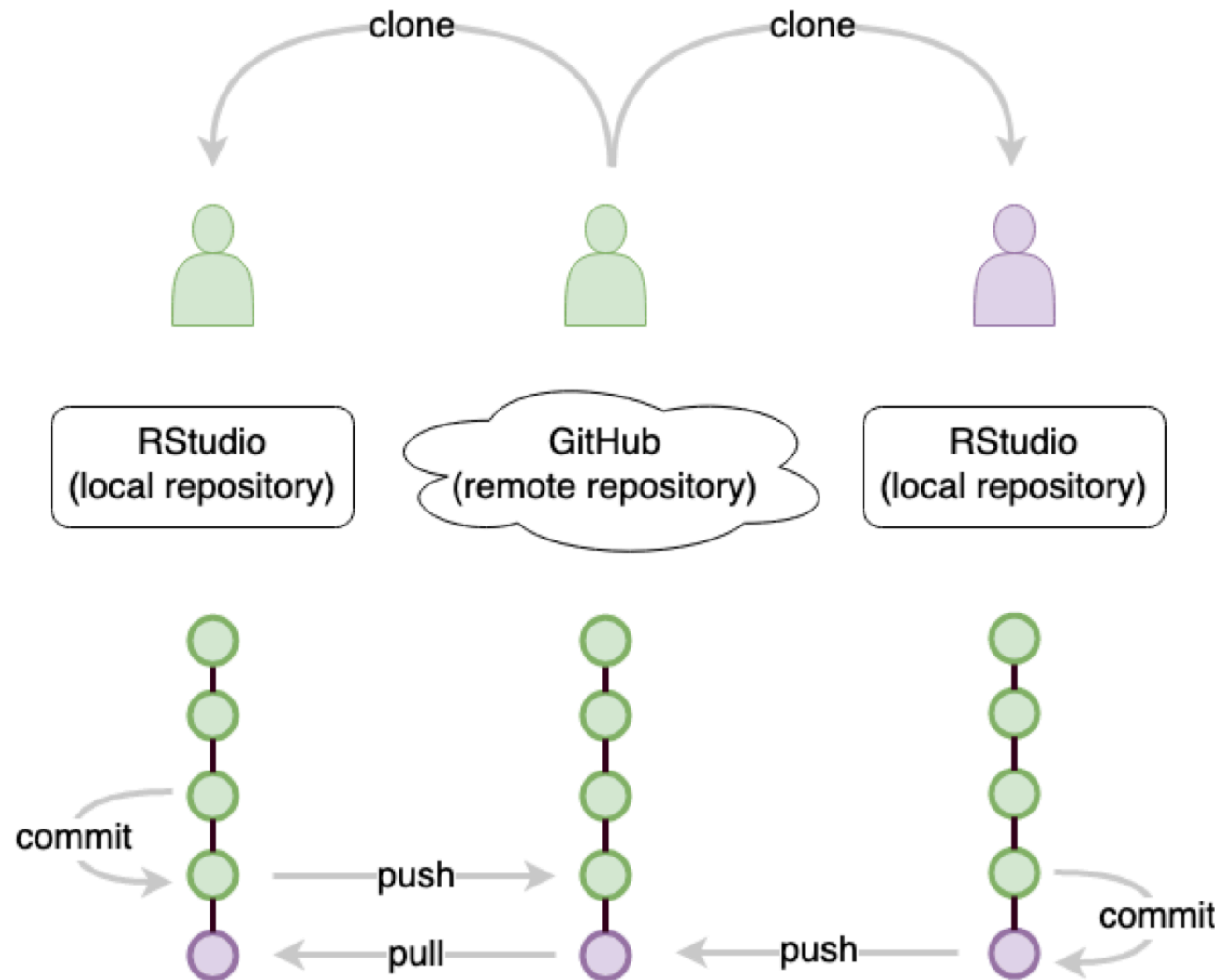
update: git push



git ???



new: git pull

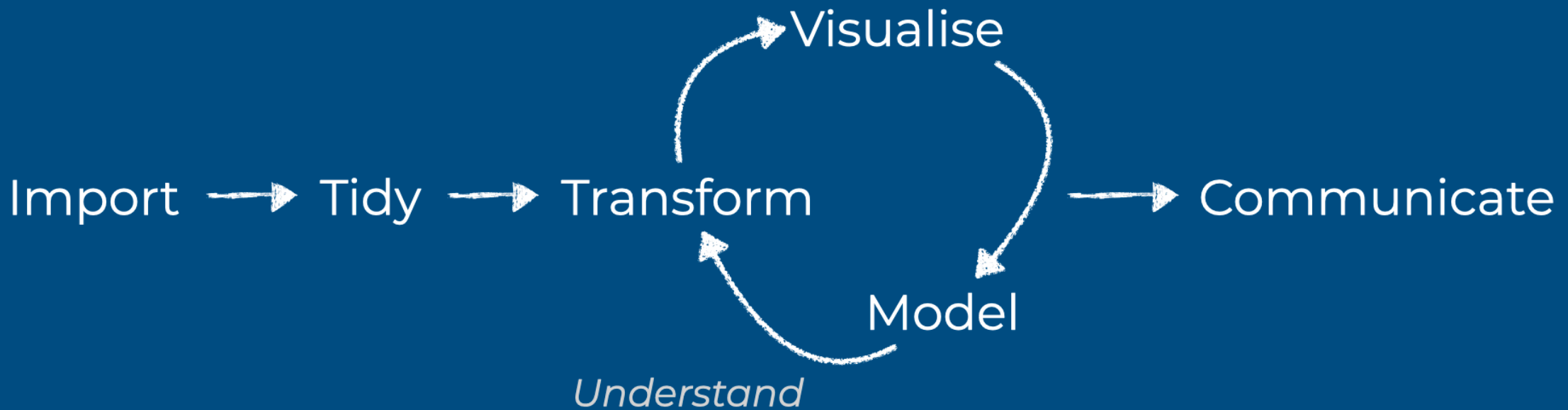


Learning Objectives (for this week)

1. Learners can list the six elements of the data science lifecycle.
2. Learners can describe the four main aesthetic mappings that can be used to visualise data using the ggplot2 R Package.
3. Learners can control the colour scaling applied to a plot using colour as an aesthetic mapping.
4. Learners can compare three different geoms (bar/col, histogram, point) and their use case.

Data Science Lifecycle

Data Science Lifecycle

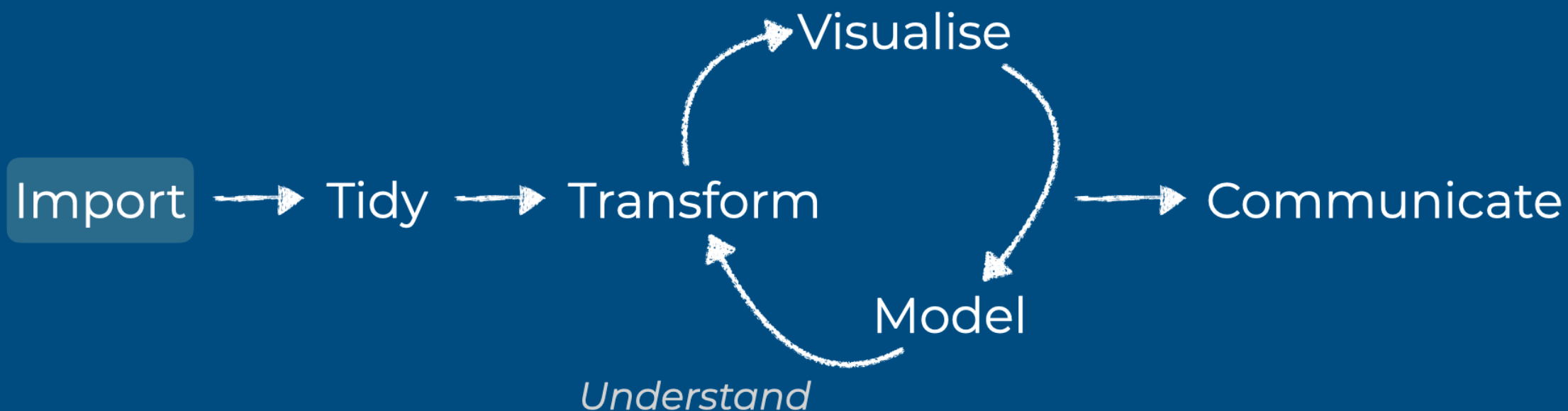


Program

From: <https://r4ds.had.co.nz/>

Data Science Lifecycle

Get your data into R

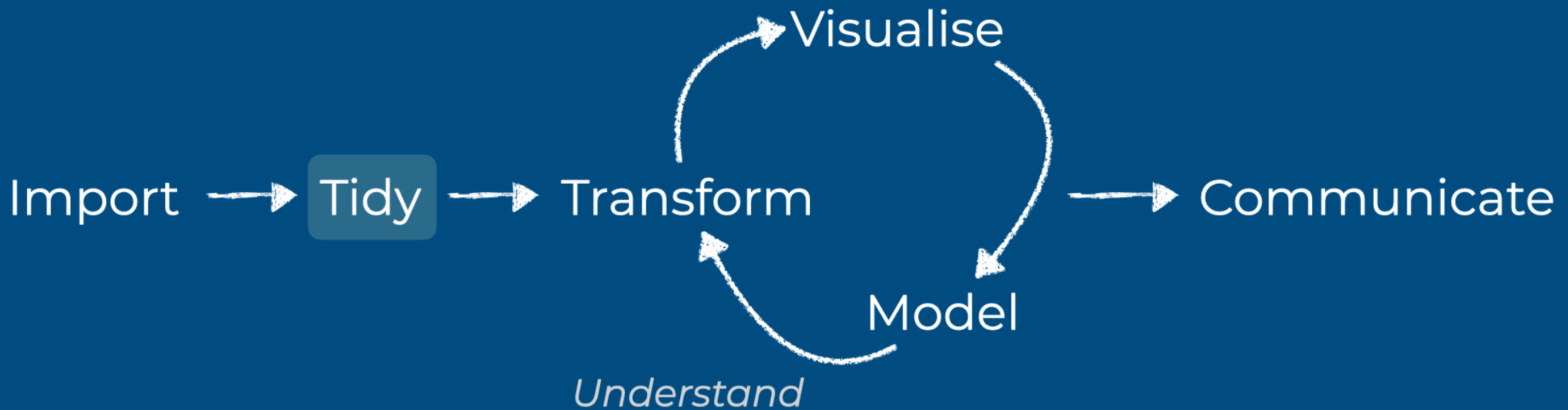


Program

From: <https://r4ds.had.co.nz/>

Data Science Lifecycle

Store your data in a consistent form

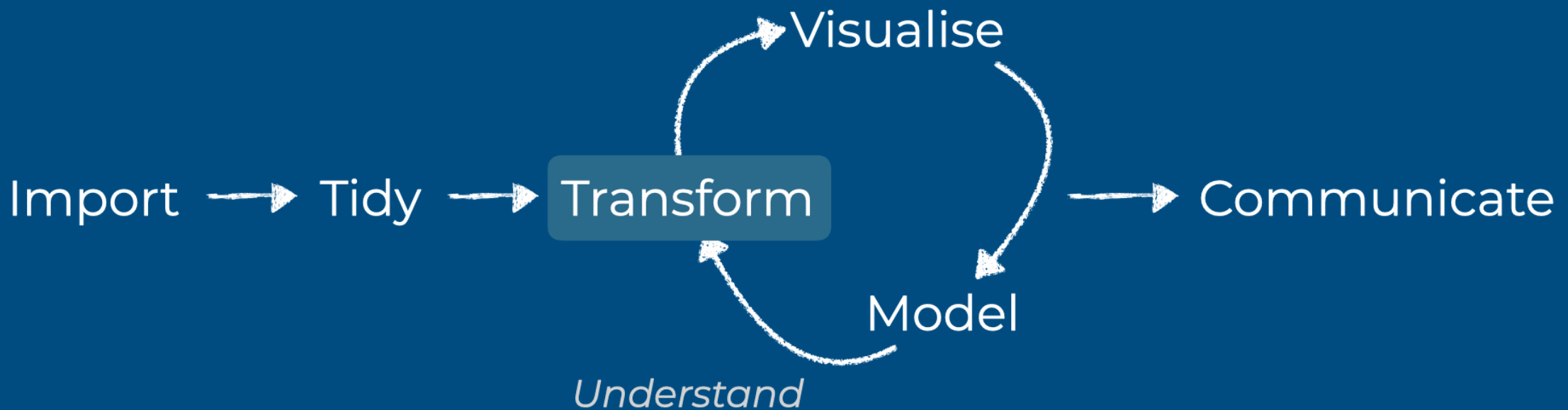


Program

From: <https://r4ds.had.co.nz/>

Data Science Lifecycle

Narrow down + Create new variables + Summary stats

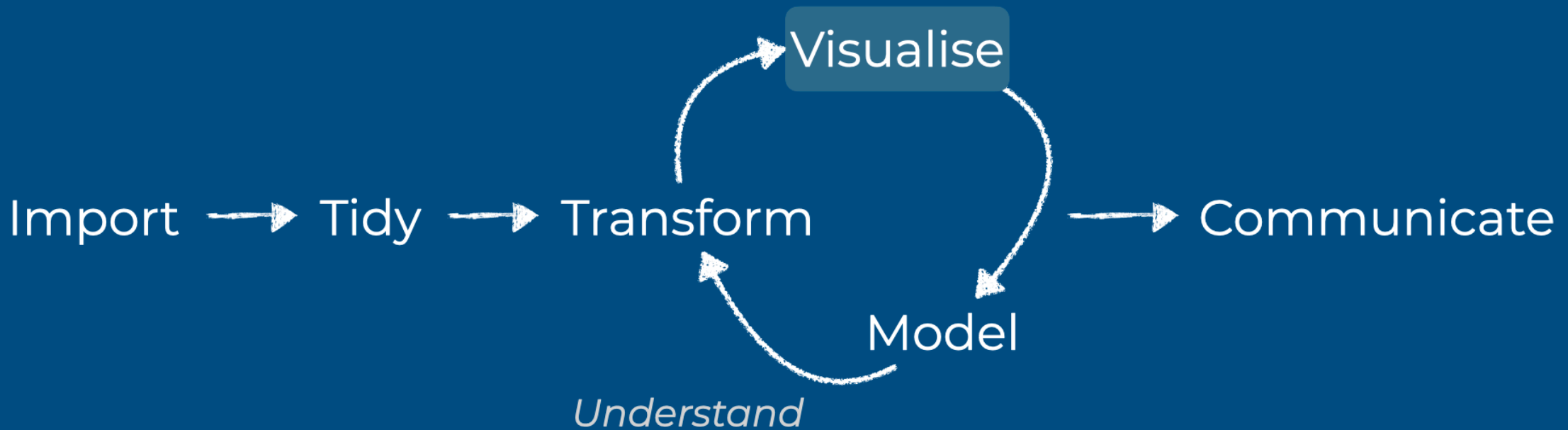


Program

From: <https://r4ds.had.co.nz/>

Data Science Lifecycle

Explore your with visual representations

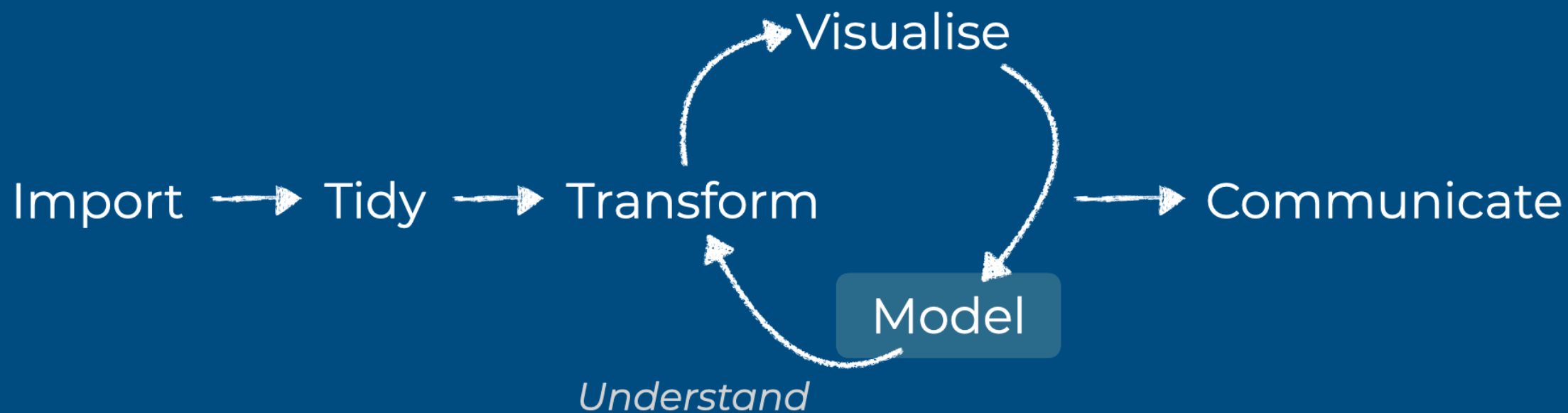


Program

From: <https://r4ds.had.co.nz/>

Data Science Lifecycle

Explore your with visual representations

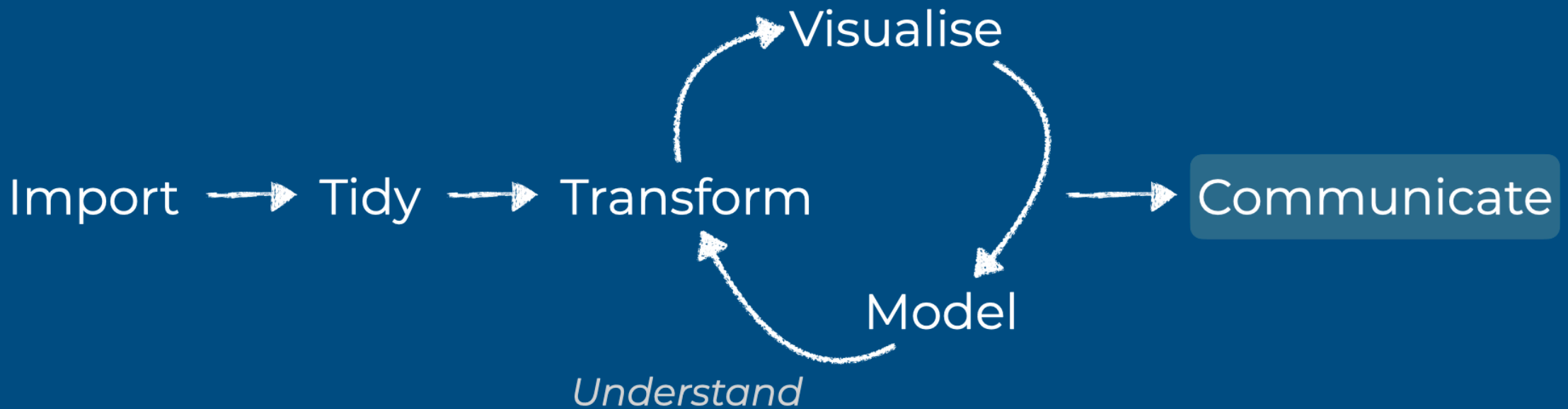


Program

From: <https://r4ds.had.co.nz/>

Data Science Lifecycle

Share your findings with others



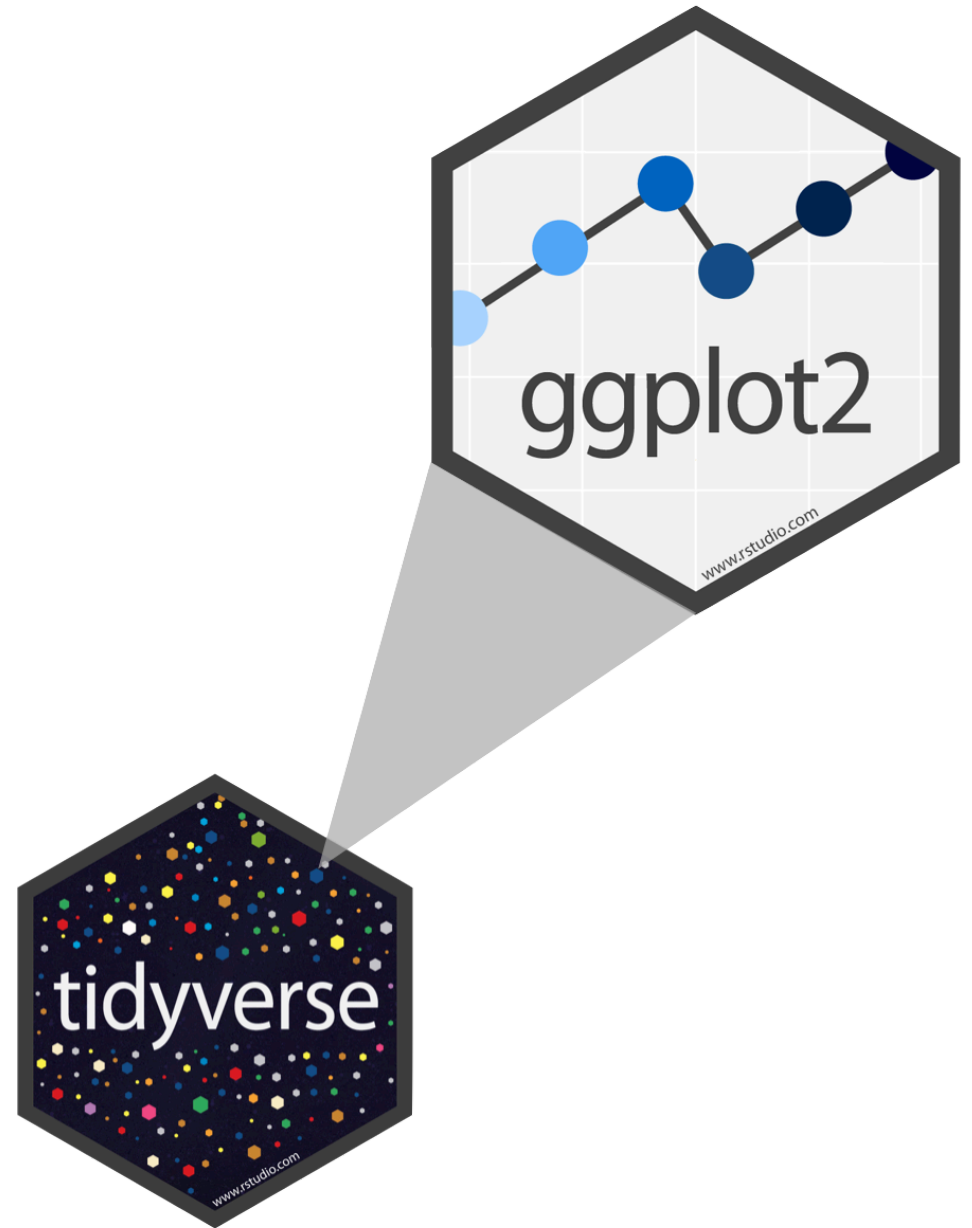
Program

From: <https://r4ds.had.co.nz/>

Exploratory Data Analysis with `ggplot2`

R Package ggplot2

- **ggplot2** is tidyverse's data visualization package
- **gg** in **ggplot2** stands for Grammar of Graphics
- Inspired by the book [Grammar of Graphics by Leland Wilkinson](#)
- **Documentation:**
<https://ggplot2.tidyverse.org/>



My turn: Working with R

Sit back and enjoy!

Take a break

Please get up and move!



Code structure

- `ggplot()` is the main function in `ggplot2`
- Plots are constructed in layers
- Structure of the code for plots can be summarized as

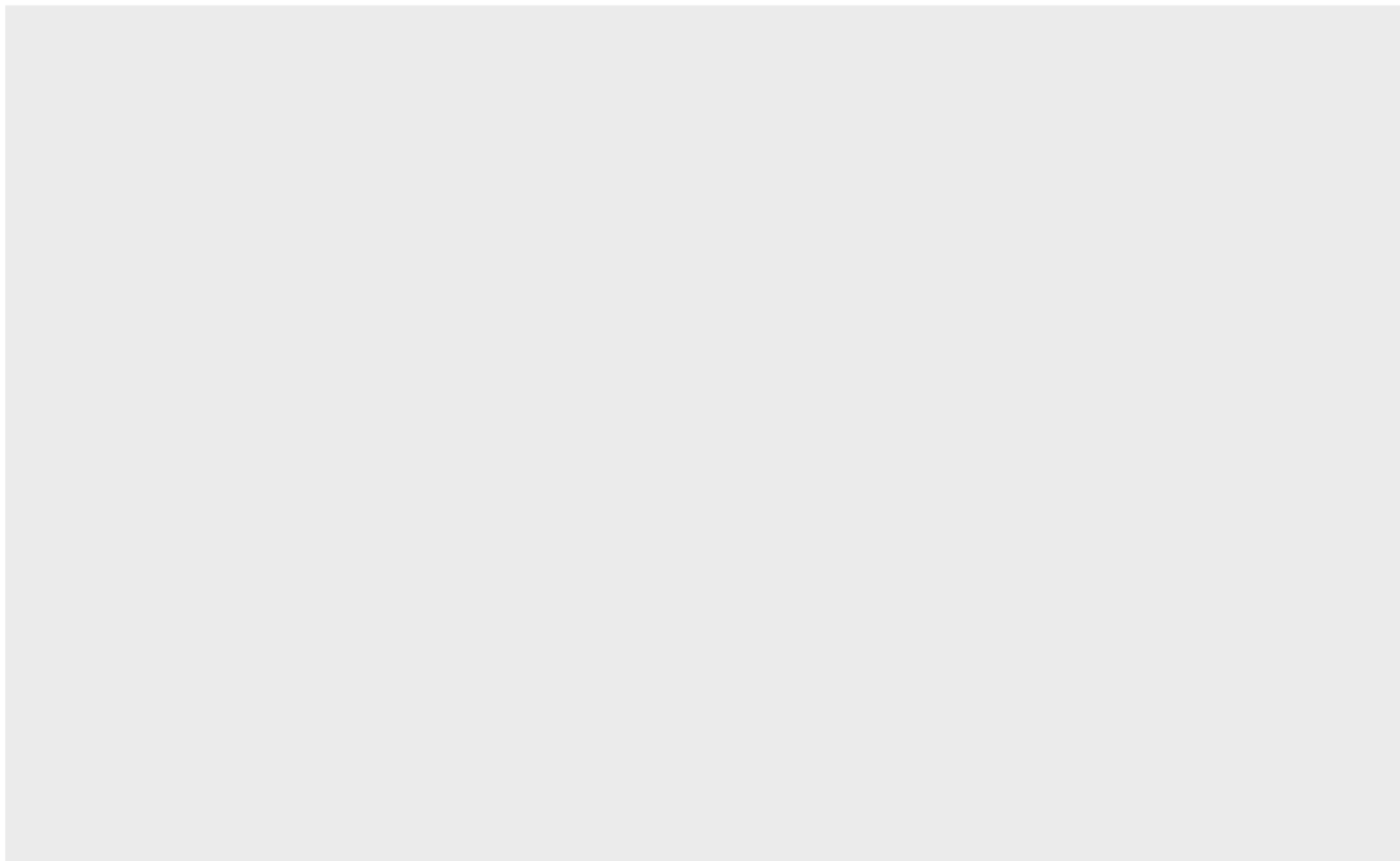
```
1 ggplot(data = [dataset],  
2       mapping = aes(x = [x-variable],  
3                     y = [y-variable])) +  
4   geom_xxx() +  
5   other options
```

Code structure

```
1 ggplot()
```

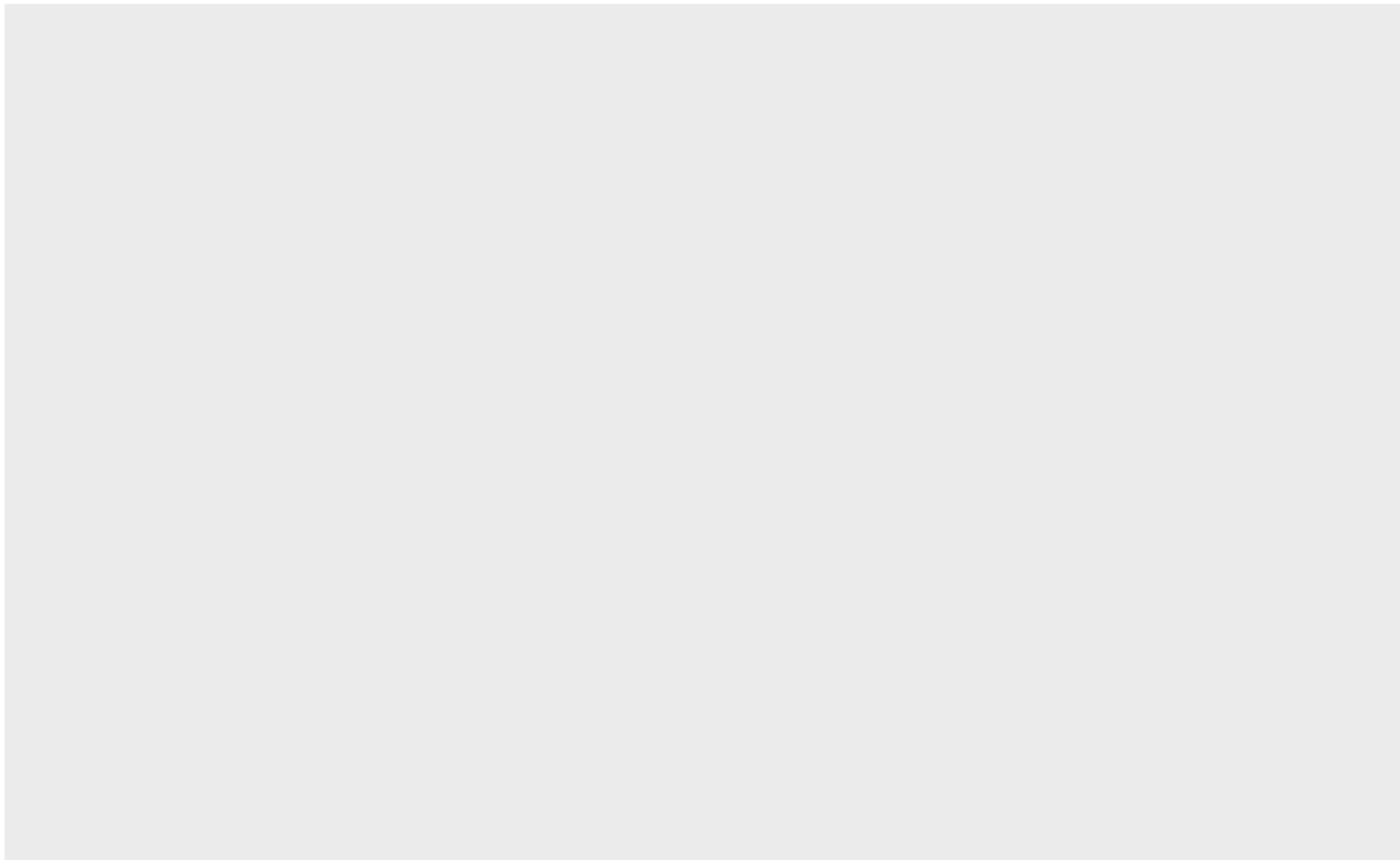
Code structure

```
1 ggplot(data = gapminder)
```



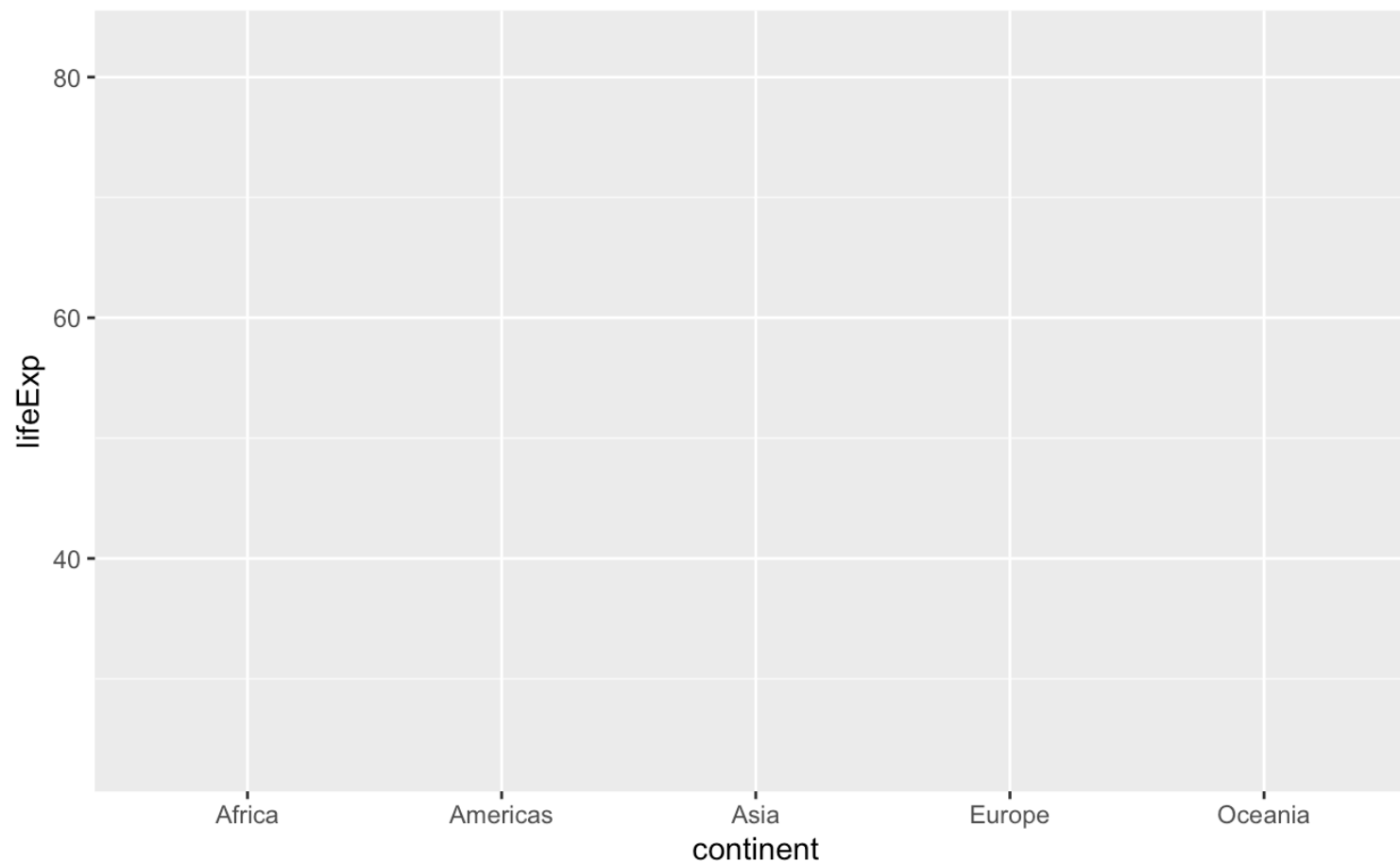
Code structure

```
1 ggplot(data = gapminder,  
2        mapping = aes())
```



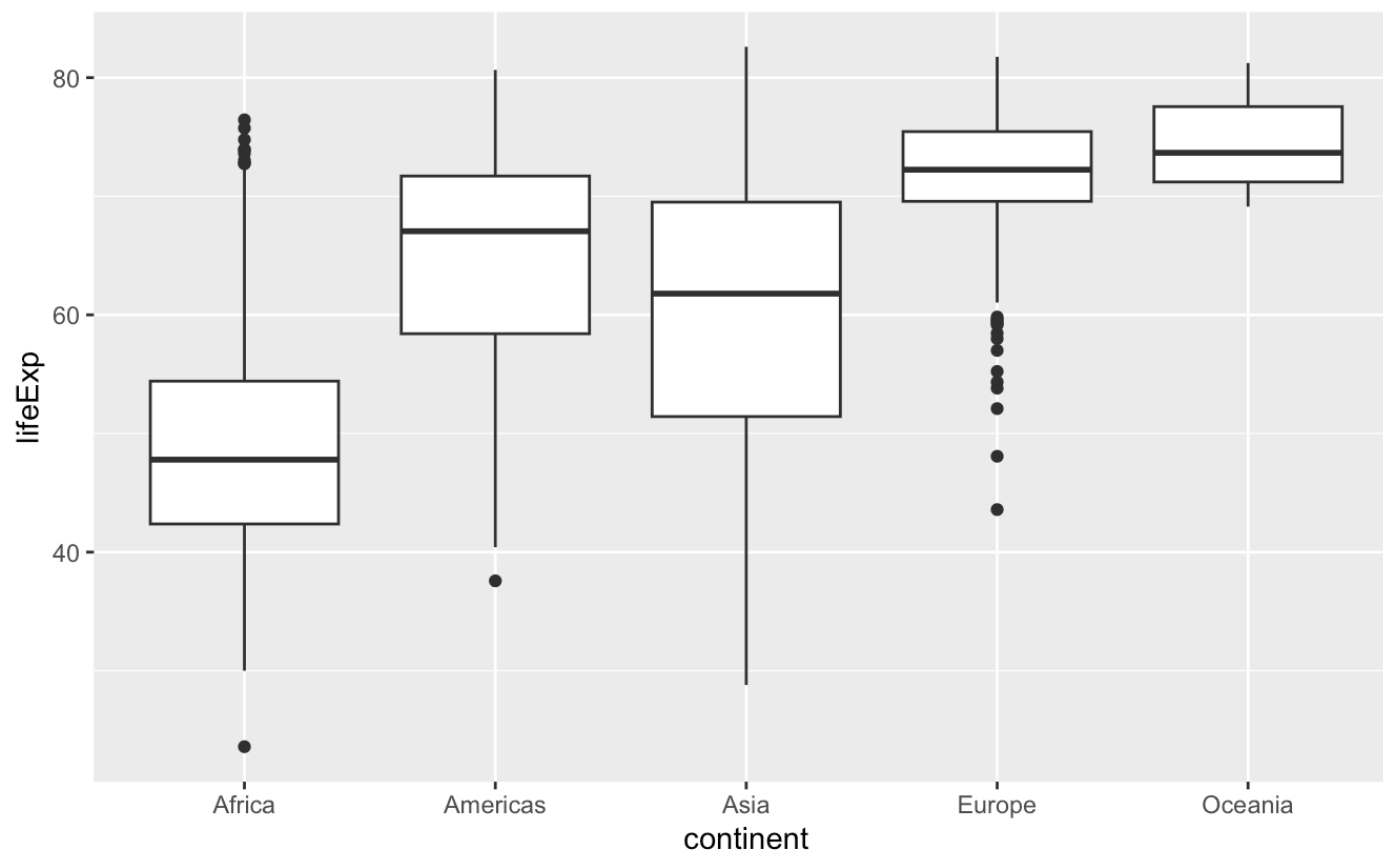
Code structure

```
1 ggplot(data = gapminder,  
2       mapping = aes(x = continent,  
3                     y = lifeExp))
```



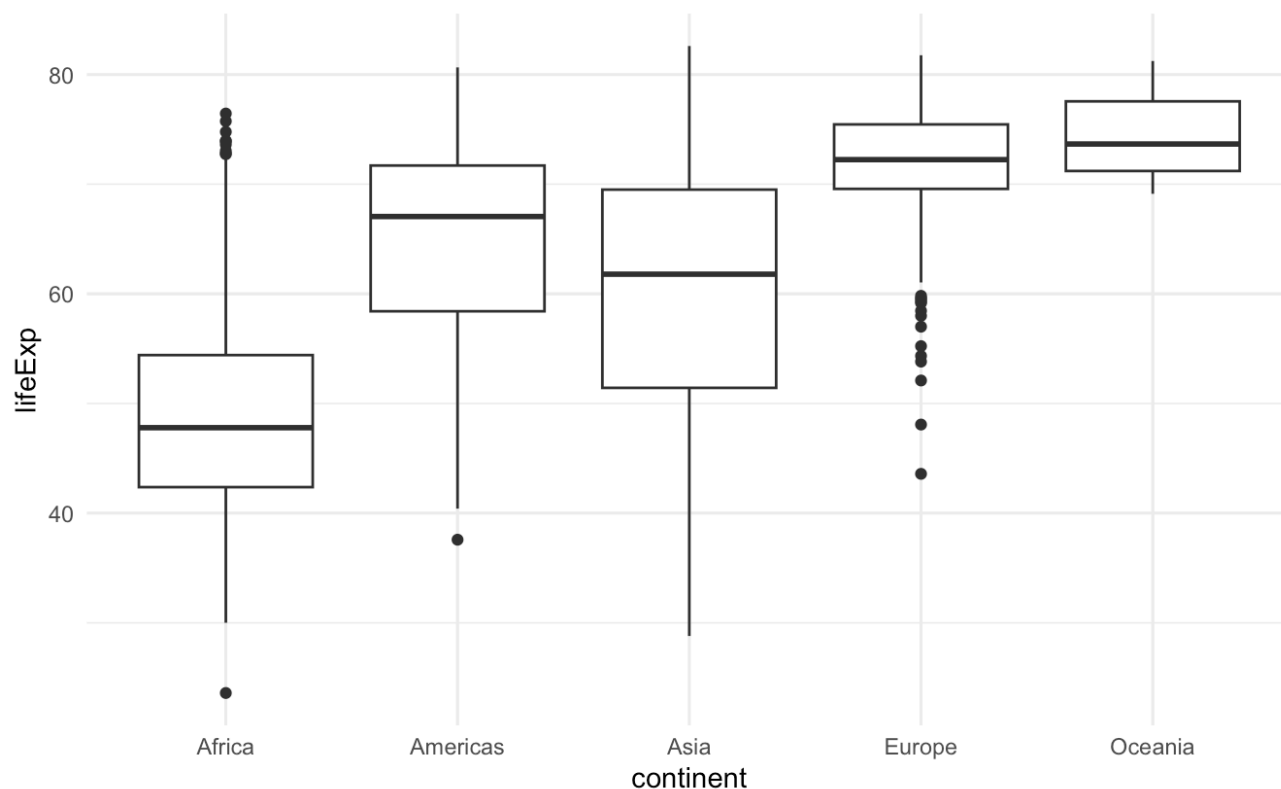
Code structure

```
1 ggplot(data = gapminder,  
2       mapping = aes(x = continent,  
3                     y = lifeExp)) +  
4   geom_boxplot()
```



Code structure

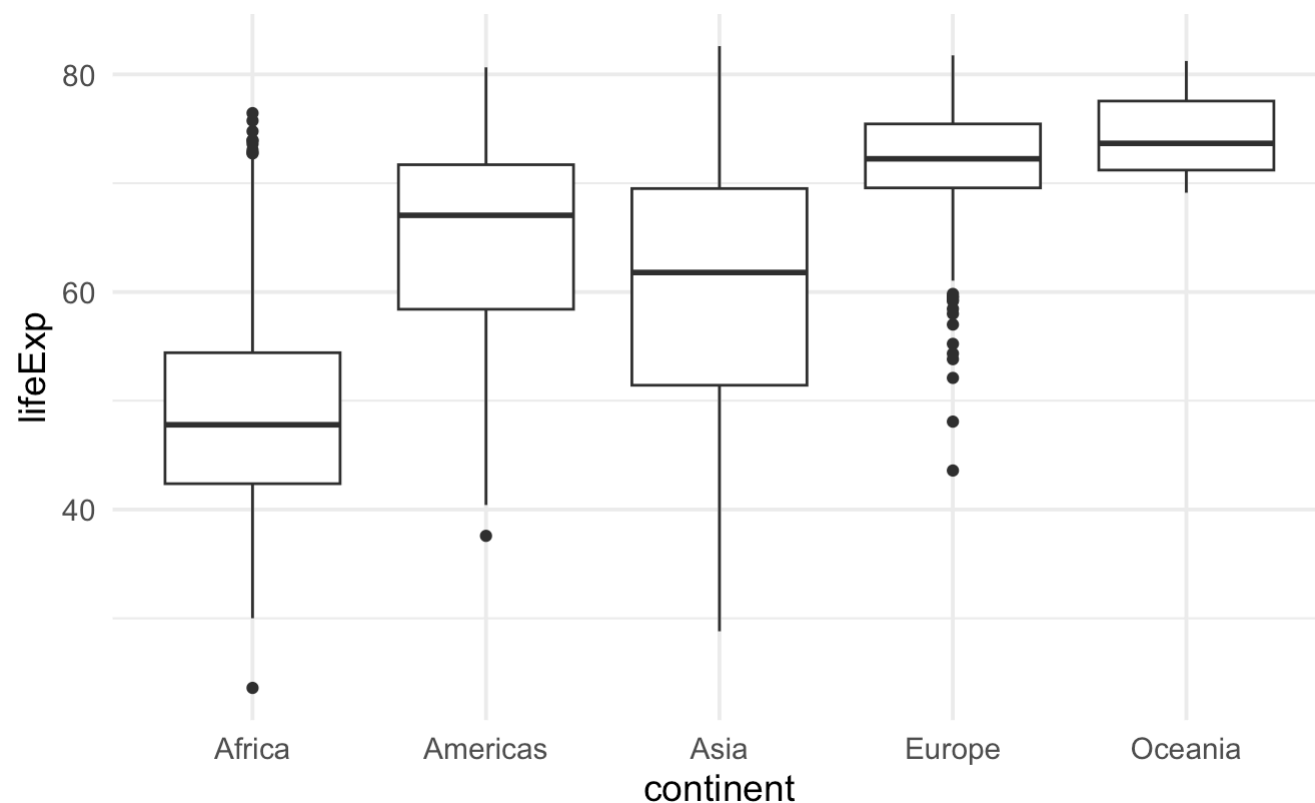
```
1 ggplot(data = gapminder,  
2       mapping = aes(x = continent,  
3                     y = lifeExp)) +  
4   geom_boxplot() +  
5   theme_minimal()
```



Polls

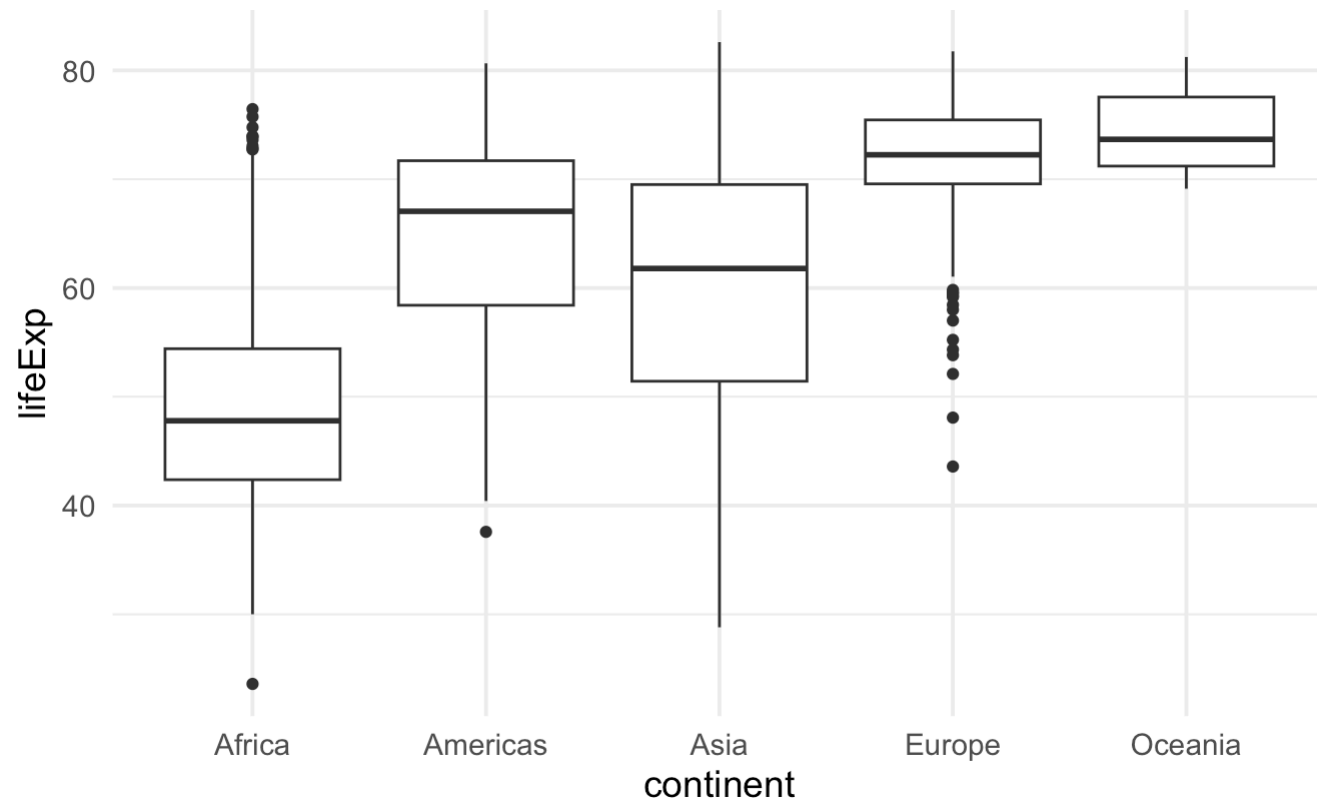
Poll 1: What does the thick line inside the box of a boxplot represent?

1. the mean of the observations
2. the middle of the box
3. the median of the observations
4. none of the above



Poll 2: What percentage of observations are contained inside the box of a boxplot (interquartile range)?

1. 25%
2. depends on the median
3. 50%
4. none of the above



Poll 3: What is the median of a set of observations?

1. The median is the most frequently occurring value in a dataset.
2. The median is the sum of all values in a dataset divided by the number of observations.
3. The median is the point above and below which half (50%) of the observations falls.
4. The median is the square root of the sum of the squares of each value in a dataset.

Boxplot, explained

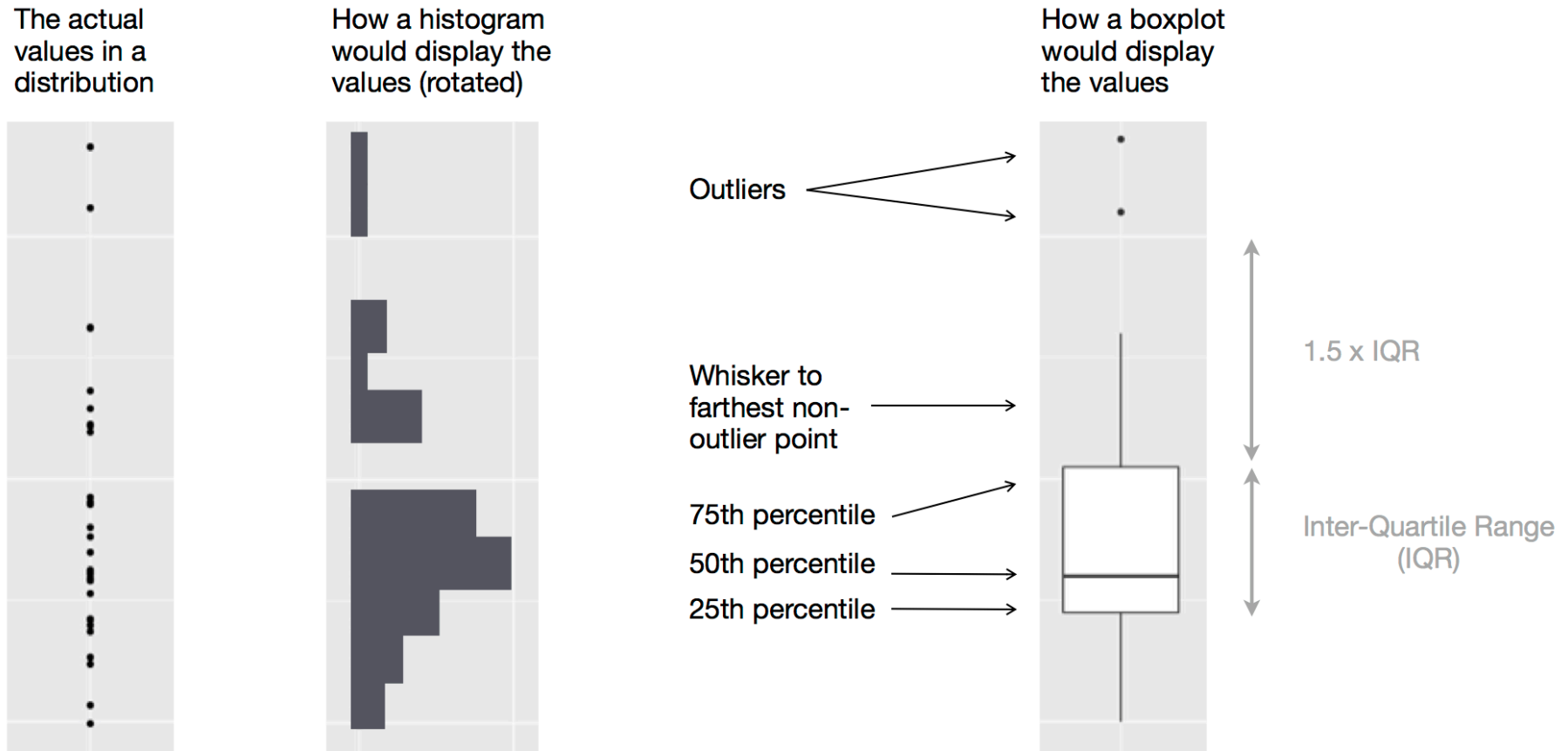


Figure 1: Diagram depicting how a boxplot is created.

Our turn: md-02-exercises

1. Open posit.cloud/spaces/663318/content in your browser.
2. Verify that the ds4owd workspace is open.
3. Click Start next to md-02-exercises.
4. In the File Manager in the bottom right window, locate the md-02b-visualize.qmd file and click on it to open it in the top left window.

Take a break

Please get up and move!



Visualizing data

Types of variables

numerical

discrete variables

- non-negative
- whole numbers
- e.g. number of students, roll of a dice

continuous variables

- infinite number of values
- also dates and times
- e.g. length, weight, size

non-numerical

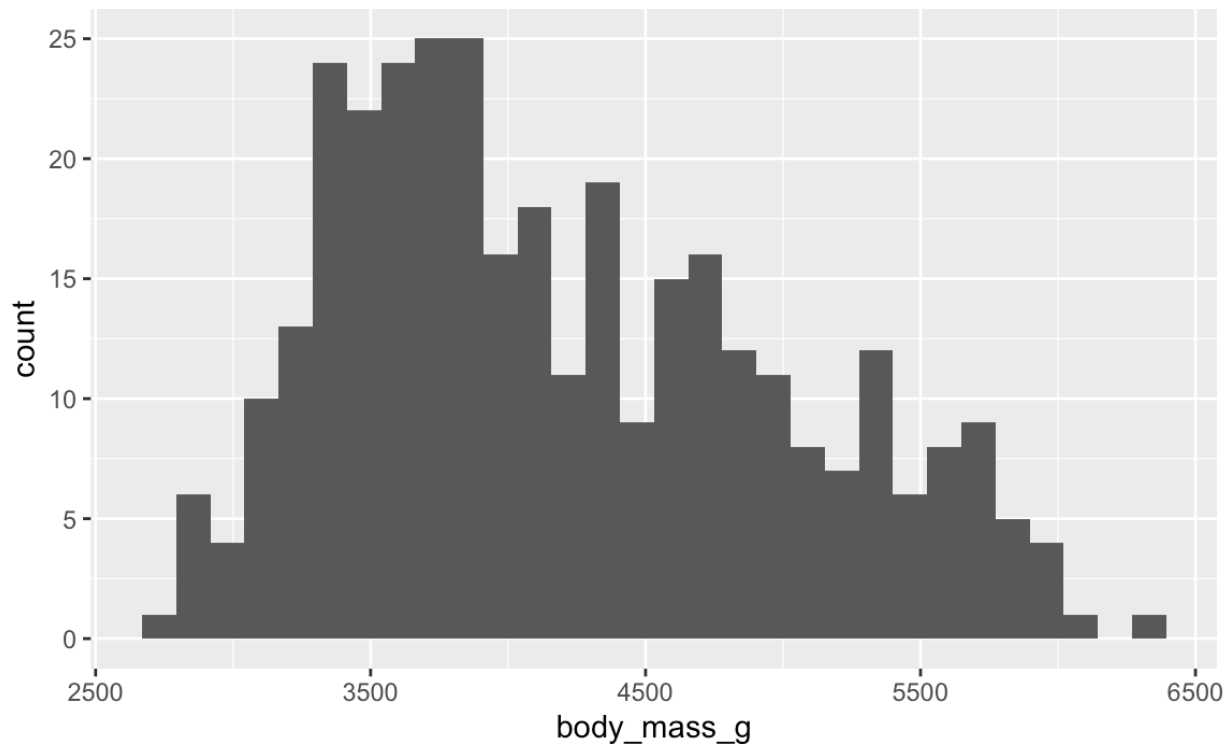
categorical variables

- finite number of values
- distinct groups (e.g. EU countries, continents)
- **ordinal** if levels have natural ordering (e.g. week days, school grades)

Histogram

- for visualizing distribution of continuous (numerical) variables

```
1 ggplot(data = penguins,  
2       mapping = aes(x = body_mass_g)) +  
3       geom_histogram()
```



Barplot

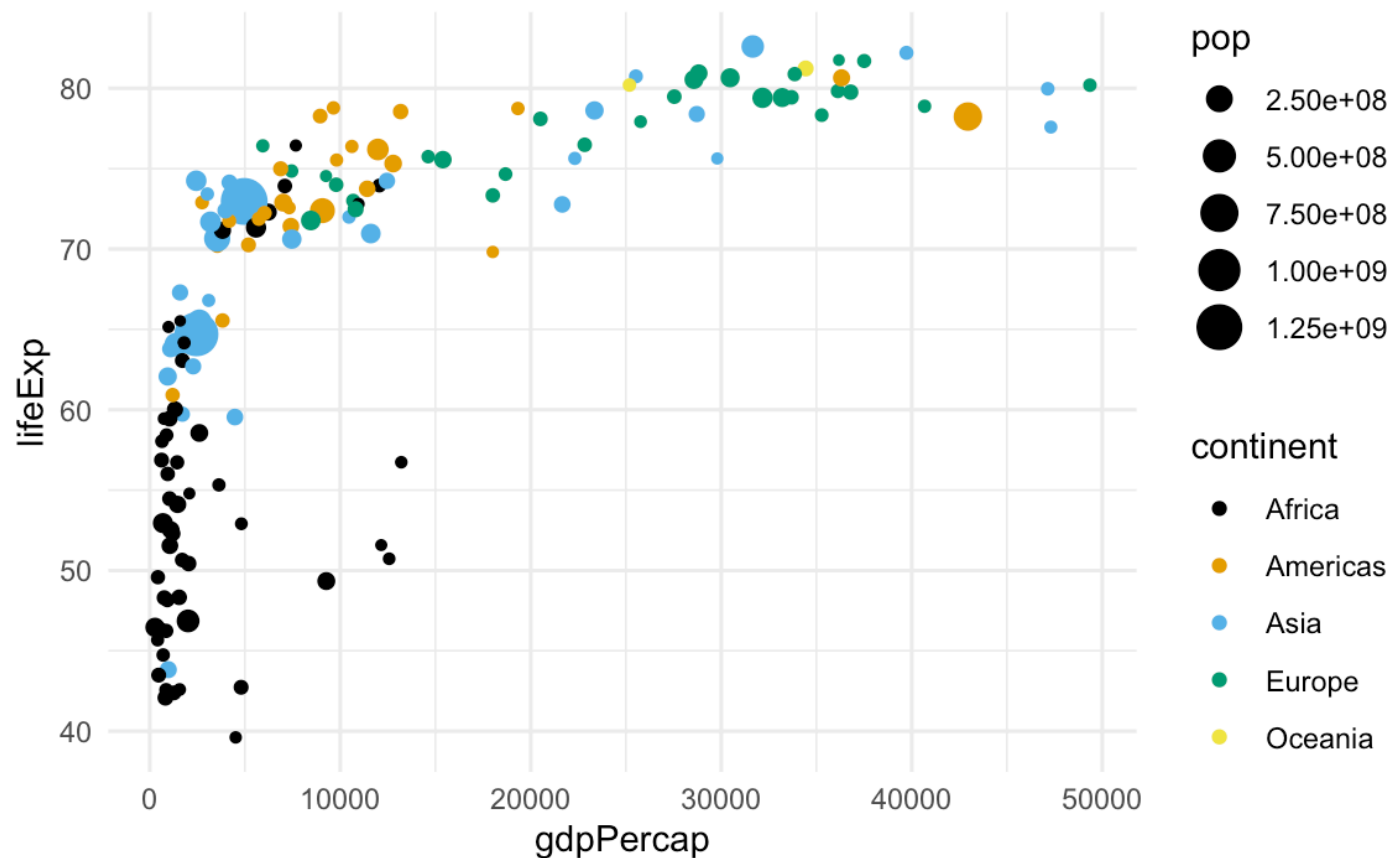
- for visualizing distribution of categorical (non-numerical) variables

```
1 ggplot(data = penguins,  
2       mapping = aes(x = species)) +  
3   geom_bar()
```

Scatterplot

- for visualizing relationships between two continuous (numerical) variables

```
1 ggplot(data = gapminder_2007,
2       mapping = aes(x = gdpPerCap,
3                     y = lifeExp,
4                     size = pop,
5                     color = continent)) +
6   geom_point() +
7   scale_color_colorblind() +
8   theme_minimal()
```

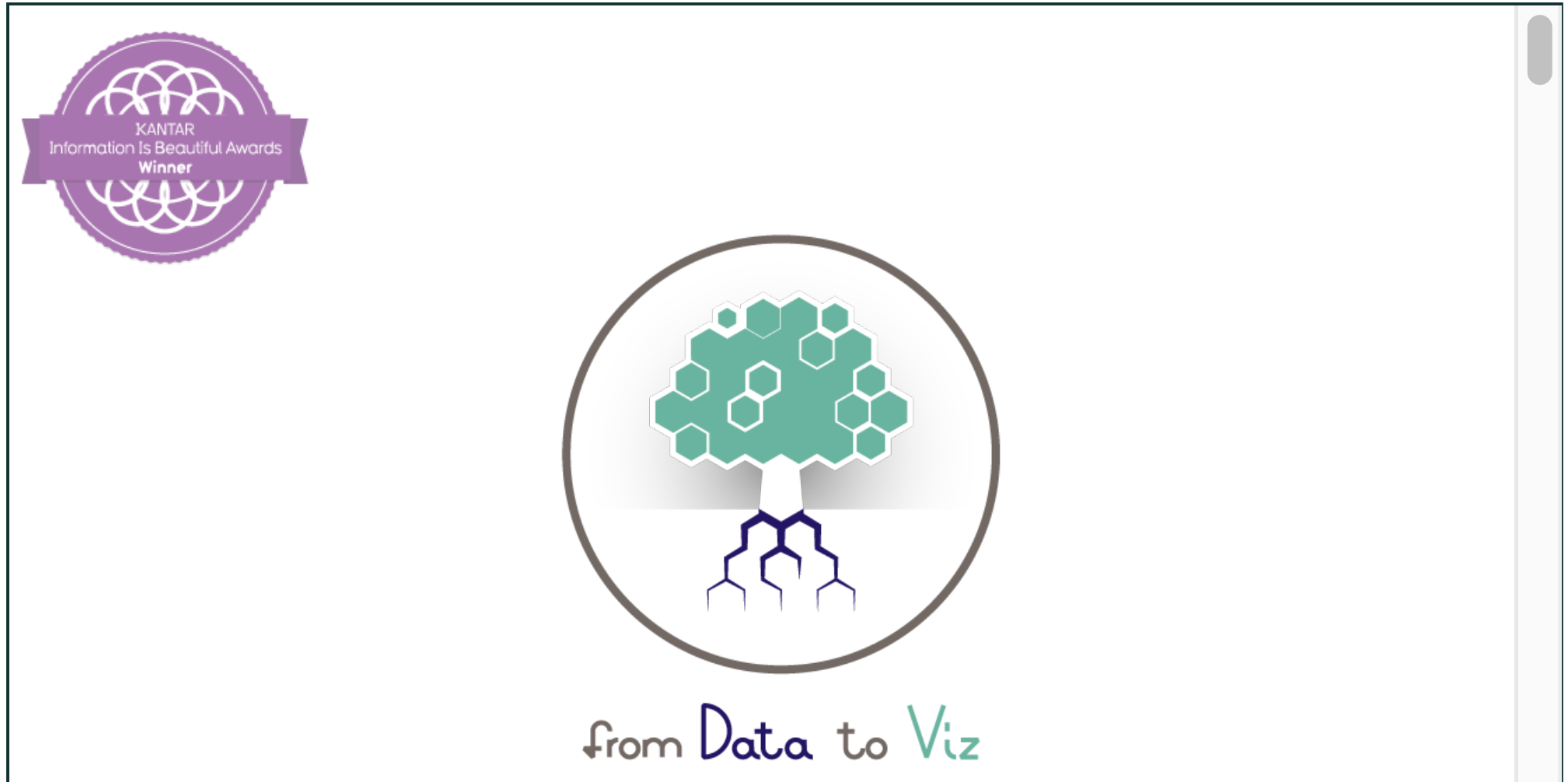


Your turn: md-02-exercises

1. If you closed it, open posit.cloud/spaces/663318/content in your browser.
2. Verify that the ds4owd workspace is open.
3. Click Continue next to md-02-exercises.
4. In the File Manager in the bottom right window, locate the md-02c-boxplot.qmd file and click on it to open it in the top left window.
5. Follow instructions in the file.

How to pick a plot?

data-to-viz.com



Homework assignments module 2

Module 2 documentation

ds4owd-002.github.io/website/modules/md-02.html

Module 2

Data science lifecycle & Exploratory data analysis using visualization

AUTHOR

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AFFILIATION

ETH Zurich

Are you ready for some data visualisations? This week is all about exploring data with `ggplot2` R package. We will also learn about the data science lifecycle.

🎯 Learning Objectives

1. Learners can list the six elements of the data science lifecycle.
2. Learners can describe the four main aesthetic mappings that can be used to visualise data using the ggplot2 R Package.
3. Learners can control the colour scaling applied to a plot using colour as an aesthetic mapping.
4. Learners can compare three different geoms (bar/col, histogram,

Homework due date

- Homework assignment due: Wednesday, 2025-09-24
- Quiz due: Wednesday, 2025-10-01

Wrap-up

Thanks!

Slides created via revealjs and Quarto:

<https://quarto.org/docs/presentations/revealjs/>

Access slides as [PDF on GitHub](#)

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